



北京大学  
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# Experimental Demonstration of Resonant Adiabatic Writing and Computing in Ferroelectric Capacitive Memory Array for Energy-Efficient Edge AI

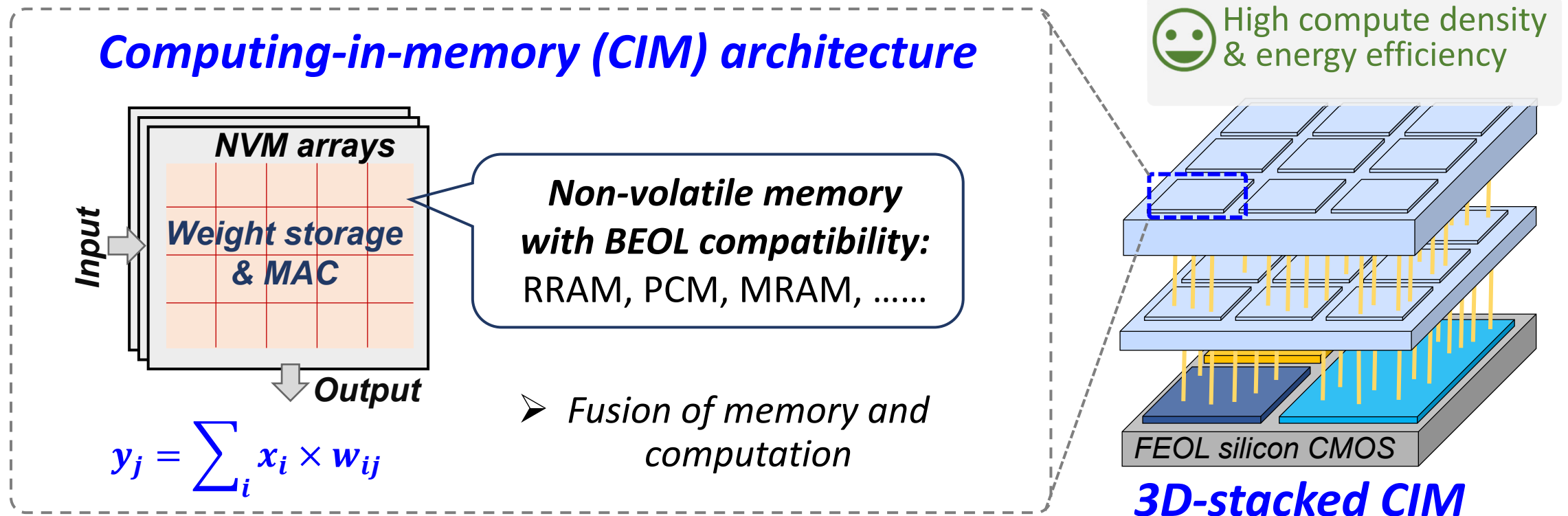
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Peking University, Beijing 100871, China*

# Motivation (1/4)

## Computing-in-memory (CIM)

- Non-volatile memory (NVM) based computing-in-memory (CIM) architecture
  - **Highly-parallel multiply-accumulate (MAC)** operation inside memory array
  - High compute density and energy efficiency with further **3D-stackable capacity**

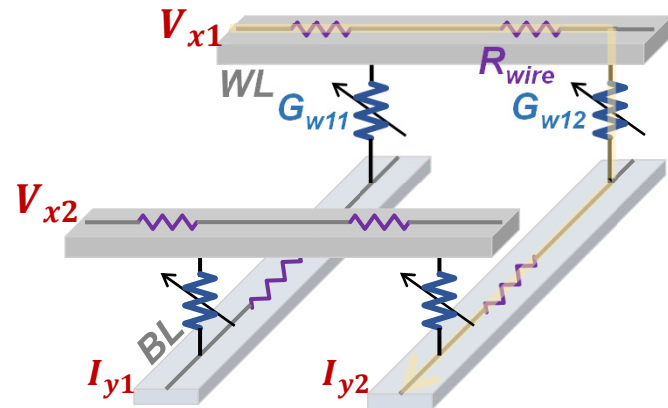


# Motivation (2/4)

## Resistive computing-in-memory (R-CIM)

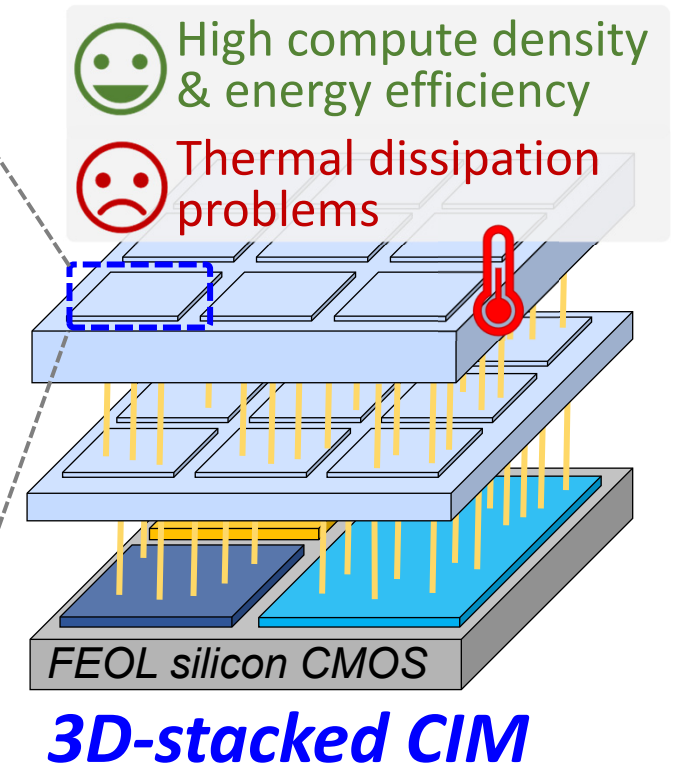
- Conventional resistive NVM-based CIM (R-CIM)
  - **Direct conduction (DC) path:** IR drop and sneak path issues
  - **Thermal dissipation problems:** Joule heating effect of current domain

### Resistive computing-in-memory (R-CIM)



$$I_{sum} = \sum V_x \cdot G_w$$

- Weight (W): *programmable resistance states*
- Current domain MAC
- High static current and static power, and inevitable thermal dissipation





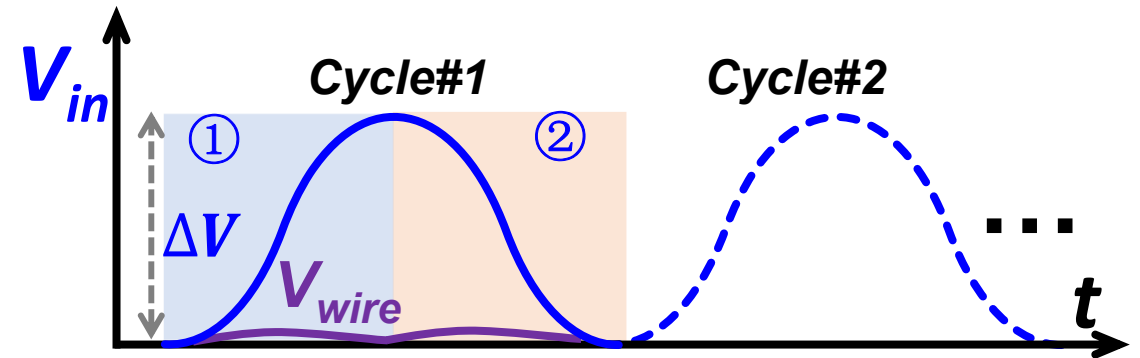
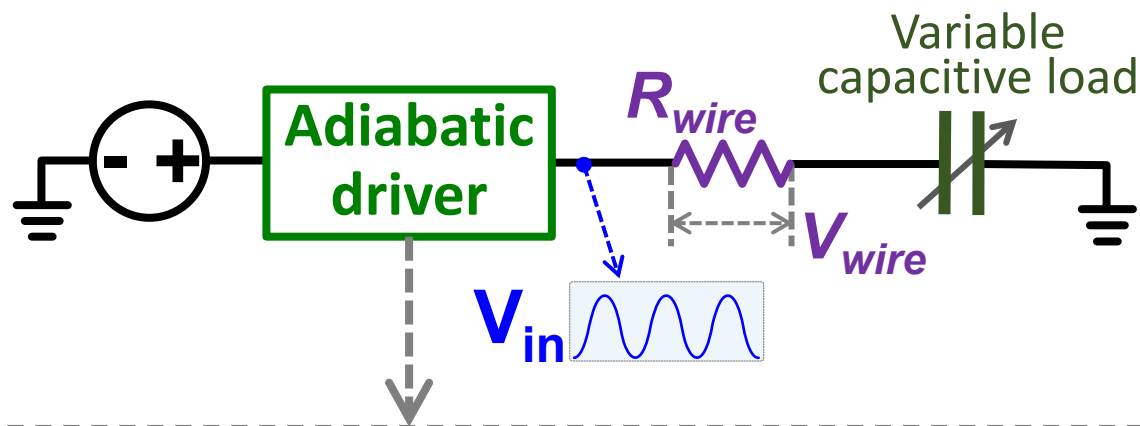
# Motivation (4/4)

## *Adiabatic technique*

- Adiabatic charge recycling for the capacitive CIM circuit

- **Adiabatic charging:** minimize resistive dissipation
- **Charge recycling:** minimize dynamic power for capacitor charging

Enable energy-efficiency improvement



① **Adiabatic charging**

Gradually ramp the supply voltage  $\rightarrow$  reduce the voltage drop across the resistive component

**Minimize resistive dissipation**

② **Charge recycling**

Reclaim the electric energy from the capacitor and transfer it back to the supply

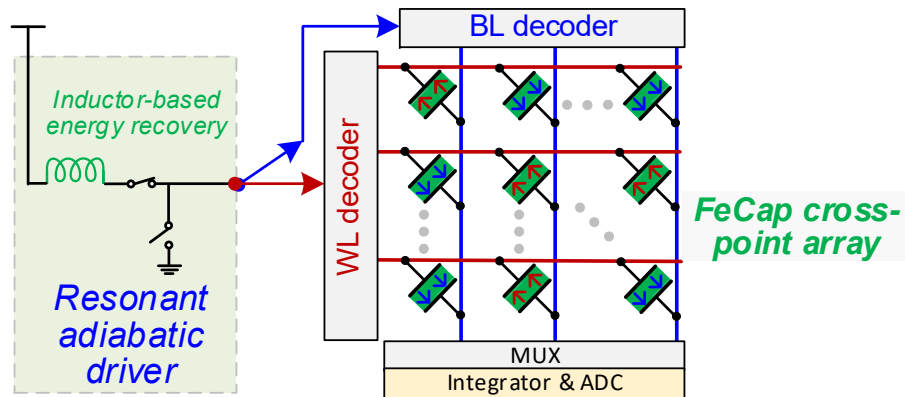
**Minimize charging energy**

$E \rightarrow 0$

# In this work of ferroelectric-based adiabatic C-CIM

- Implement **resonant adiabatic (RA)** writing and computing in C-CIM based on **3D stackable ferroelectric (FE) capacitive memory array**, achieving high energy efficiency

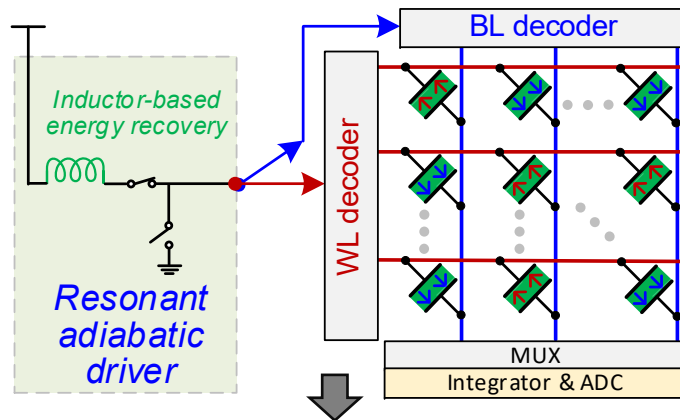
## *3D stackable FeCap based resonant adiabatic C-CIM*



# In this work of ferroelectric-based adiabatic C-CIM

- Implement **resonant adiabatic (RA)** writing and computing in C-CIM based on **3D stackable ferroelectric (FE) capacitive memory array**, achieving high energy efficiency

## 3D stackable FeCap based resonant adiabatic C-CIM



## Ferroelectric capacitor (FeCap) array based RA C-CIM

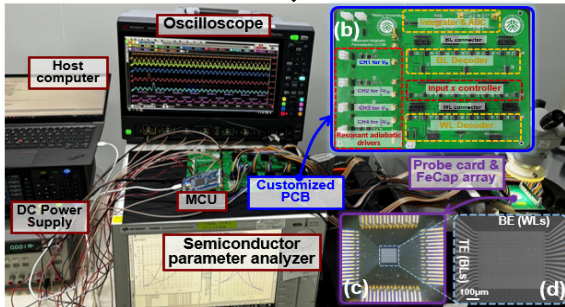
*FE polarization modulated capacitance states*

**Capacitive weight storage**

*Capacitive property of both write and read operation*

**Adiabatic charging & resonant energy recycling**

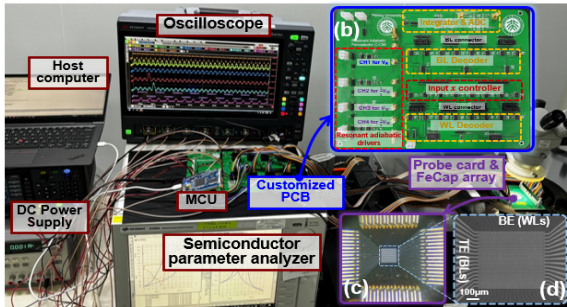
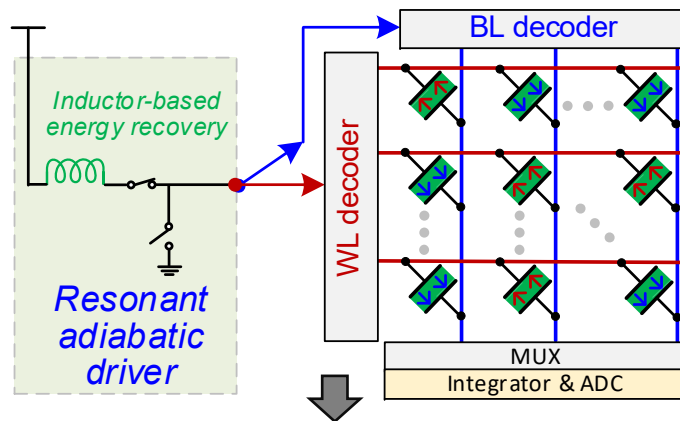
- ✓ Experimentally construct FeCap array-based RA C-CIM circuit system
- ✓ **RA writing and computing** with **complementary capacitive weight**
- ✓ Measured power consumption demonstrates a significant advantage



# In this work of ferroelectric-based adiabatic C-CIM

- Implement **resonant adiabatic (RA)** writing and computing in C-CIM based on **3D stackable ferroelectric (FE) capacitive memory array**, achieving high energy efficiency

## 3D stackable FeCap based resonant adiabatic C-CIM



## Ferroelectric capacitor (FeCap) array based RA C-CIM

*FE polarization modulated capacitance states*

**Capacitive weight storage**

*Capacitive property of both write and read operation*

**Adiabatic charging & resonant energy recycling**

- ✓ Experimentally construct FeCap array-based RA C-CIM circuit system
- ✓ **RA writing and computing** with **complementary capacitive weight**
- ✓ Measured power consumption demonstrates a significant advantage

## FeCap-based resonant adiabatic C-CIM for edge AI

- ✓ Co-design with algorithm, enabling high-accuracy inference
- ✓ High energy efficiency for both weight update and computation



# Outline

- Design of FeCap array-based resonant adiabatic (RA) C-CIM
- Resonant adiabatic writing and computing operation
- FeCap-based RA C-CIM macro for edge application
- Summary

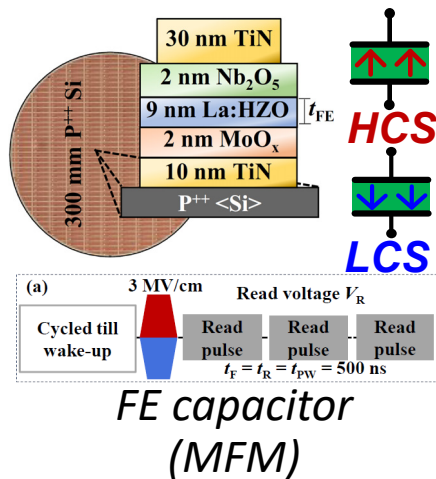
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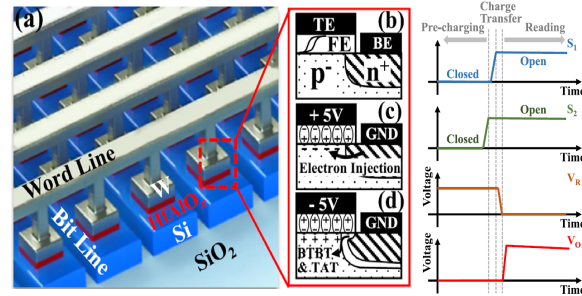
# Ferroelectric (FE) capacitive NVM for C-CIM

- Emerging **ferroelectric (FE)-based capacitive NVM** for C-CIM
  - Utilize non-volatile FE polarization to modulate capacitance states

## Conventional ferroelectric-based C-CIM



S. Mukherjee, et al., IEDM 2023

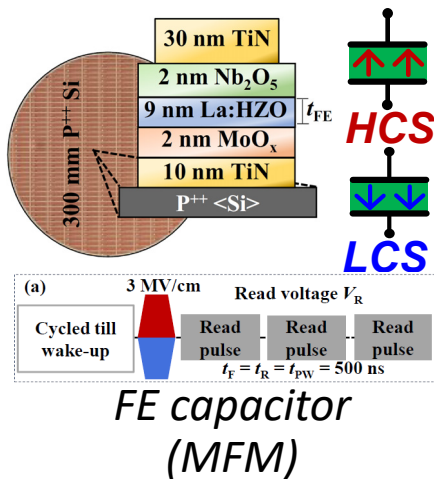


Z. Zhou, et al., VLSI 2023

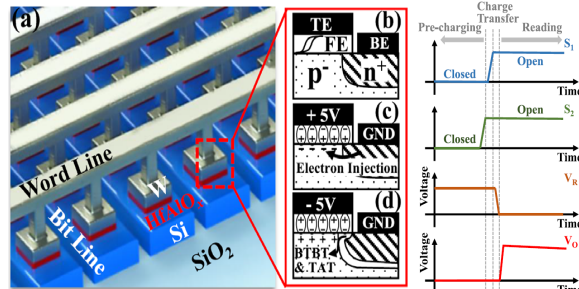
# Ferroelectric (FE) capacitive NVM for C-CIM

- Emerging **ferroelectric (FE)-based capacitive NVM** for C-CIM
  - Utilize non-volatile FE polarization to modulate capacitance states
- Conventional pulse-based charging scheme** for writing and computing
  - Dominant dynamic power consumption → **Still hinder energy efficiency**

## Conventional ferroelectric-based C-CIM



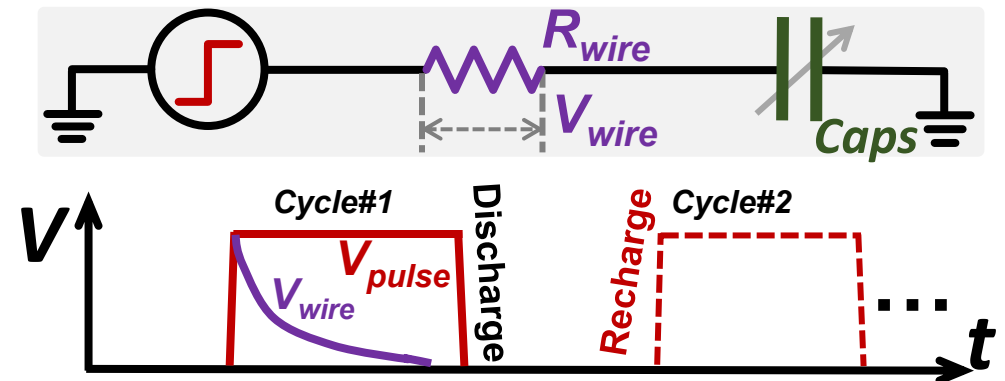
S. Mukherjee, et al., IEDM 2023



FE modulated MOS capacitor (MFSM)

Z. Zhou, et al., VLSI 2023

## Pulse-based writing and computing

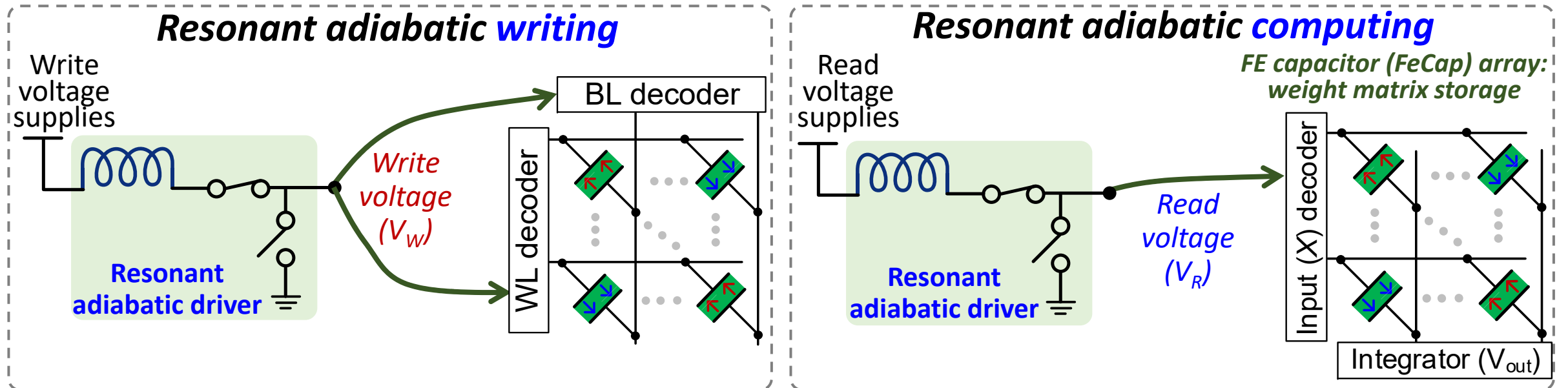


$$\text{Energy consumption } E = \frac{1}{2} C \cdot V^2 \text{ (1)} + \frac{1}{2} C \cdot V^2 \text{ (2)}$$

① Charging caps ② Joule heating in resistive component

# FeCap array-based resonant adiabatic C-CIM (1/4)

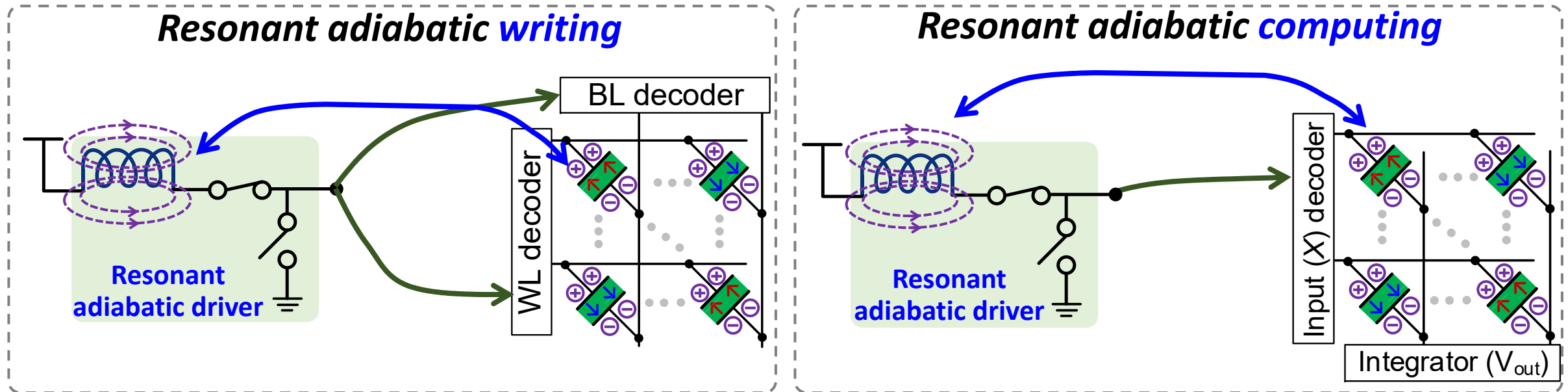
- **This work:** resonant adiabatic (RA) C-CIM based on FeCap array
  - **Resonant adiabatic driver:**
    - Incorporate an inductor ( $L$ ) for **resonant energy recycling** and **adiabatic charging**



- ✓ Generate voltage for write ( $V_w$ ) and read ( $V_R$ ) by resonance between an inductor ( $L$ ) and FeCap array
- ✓ Decoders of word-lines (WLs) and bit-lines (BLs): route  $V_w$  and  $V_R$  for writing and computing, respectively

# FeCap array-based resonant adiabatic C-CIM (2/4)

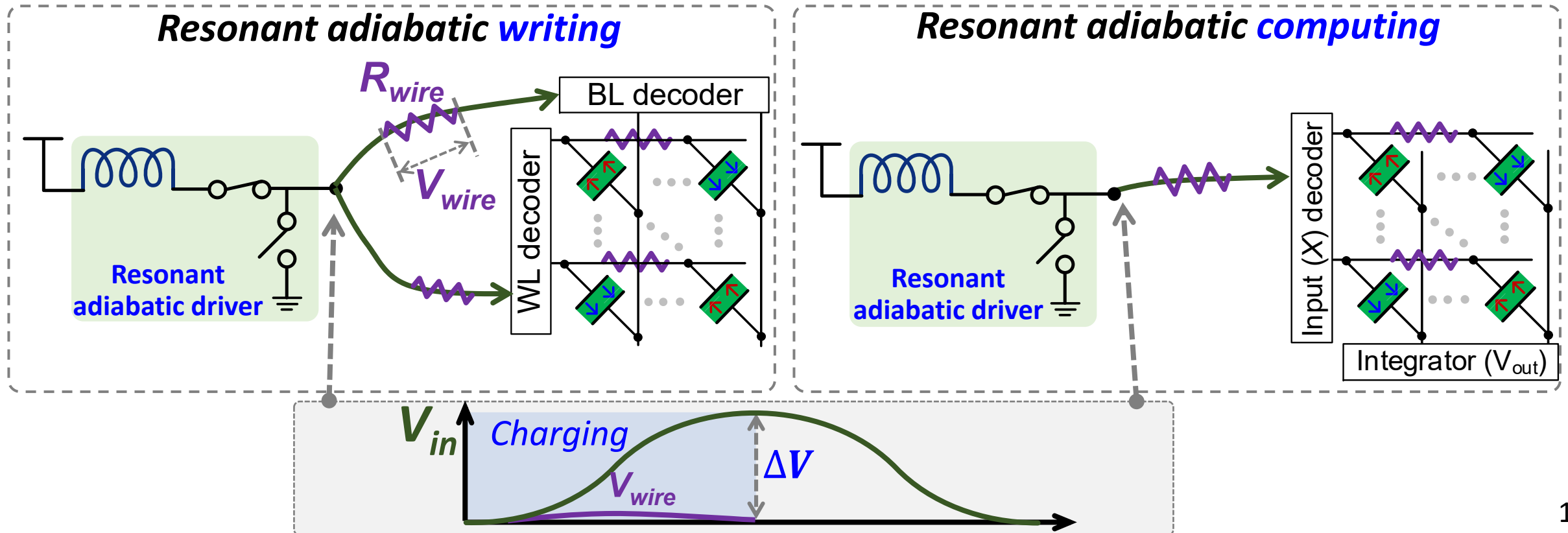
- Resonant energy recycling
  - Energy alternates between **magnetic energy in inductor** and **electric energy in capacitors**



The energy used to charge FeCaps can be transferred to inductor after one write (/compute) operation, subsequently reused for the next operation cycle.

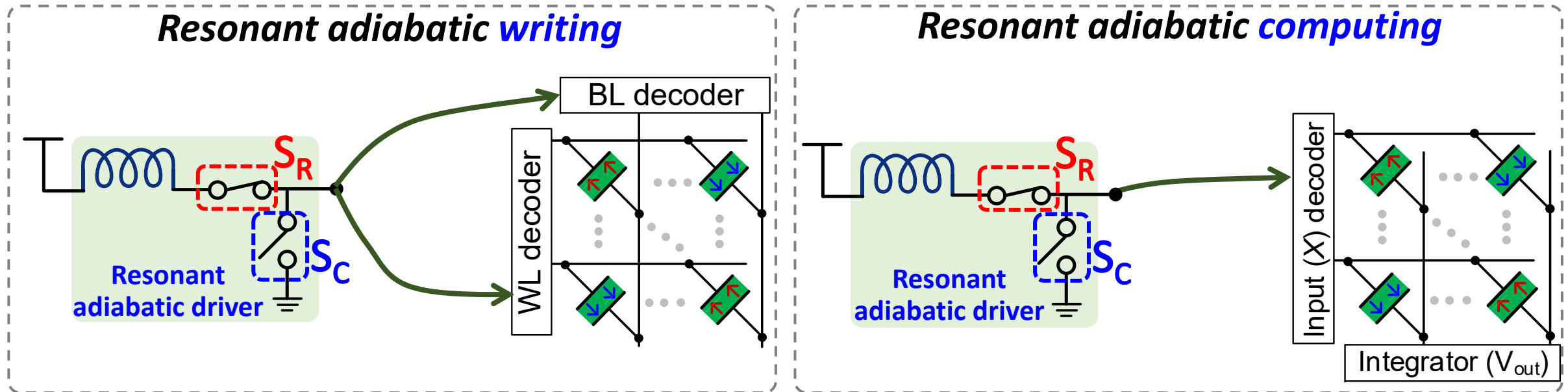
# FeCap array-based resonant adiabatic C-CIM (3/4)

- Adiabatic charging
  - **Ramping harmonic voltage**  $\rightarrow$  reduced Joule heating dissipation in resistive components



# FeCap array-based resonant adiabatic C-CIM (4/4)

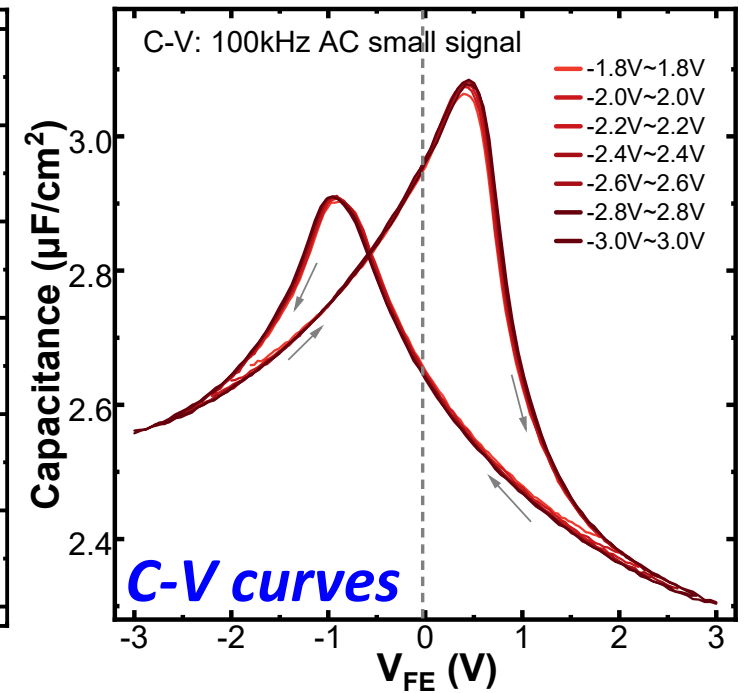
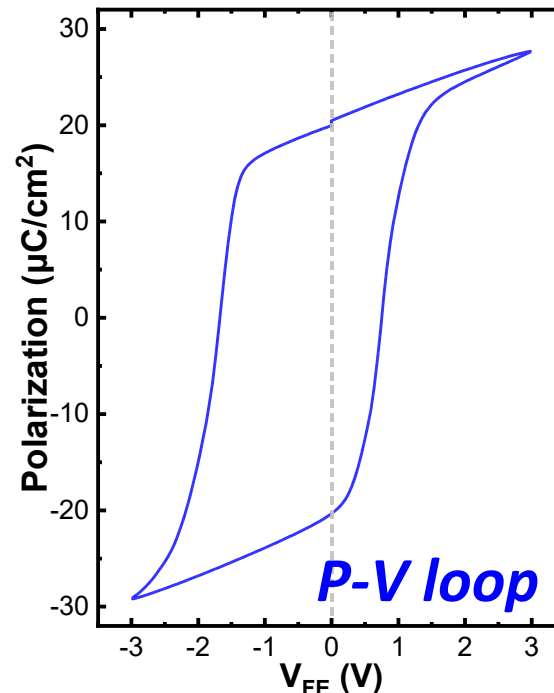
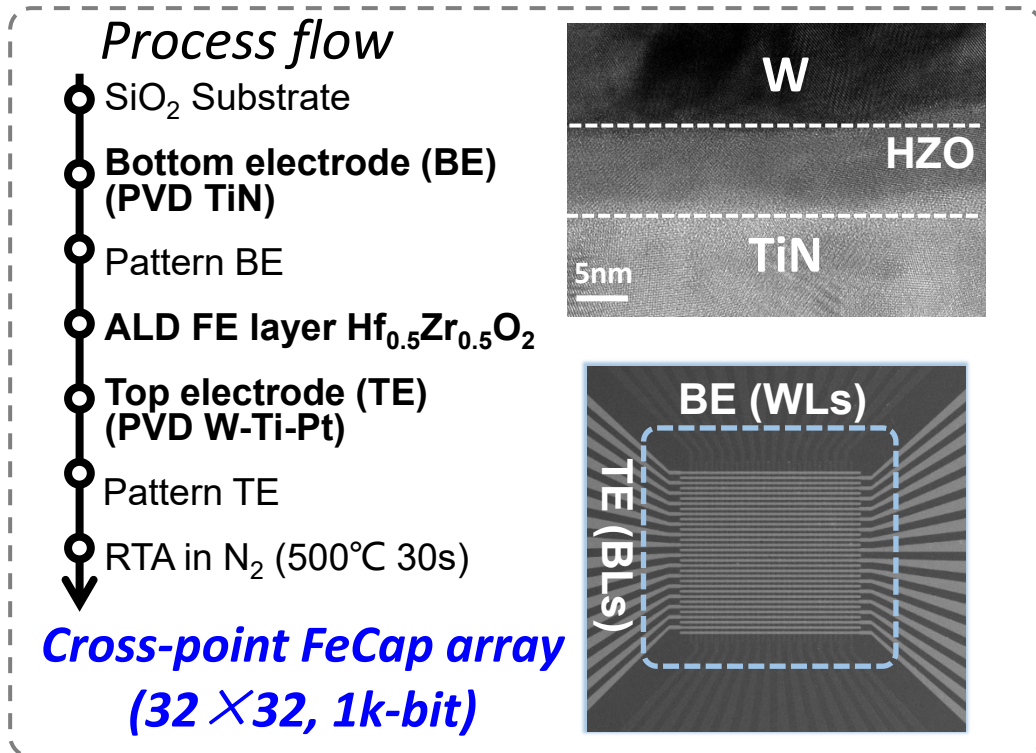
- Resonance rhythm control
  - Two anti-phase switches for resonance connection ( $S_R$ ) and dissipated energy compensation ( $S_C$ )





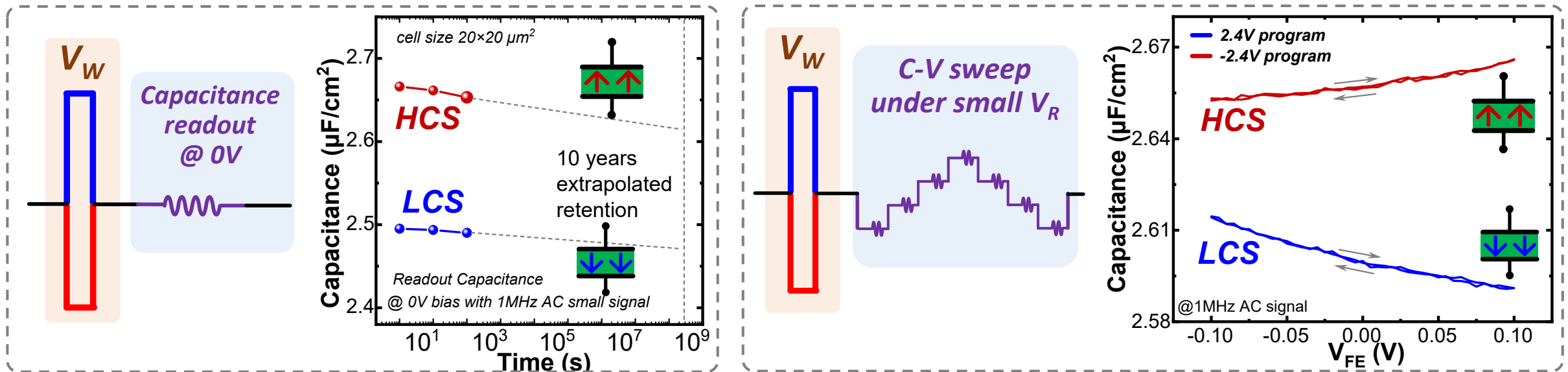
# Characteristics of FeCap array (1/2)

- Fabrication of ferroelectric capacitor (FeCap) array
  - Cross-point structure with 10nm  $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$  ferroelectric (FE) layer
  - Asymmetric small signal C-V curves due to the asymmetric electrode design



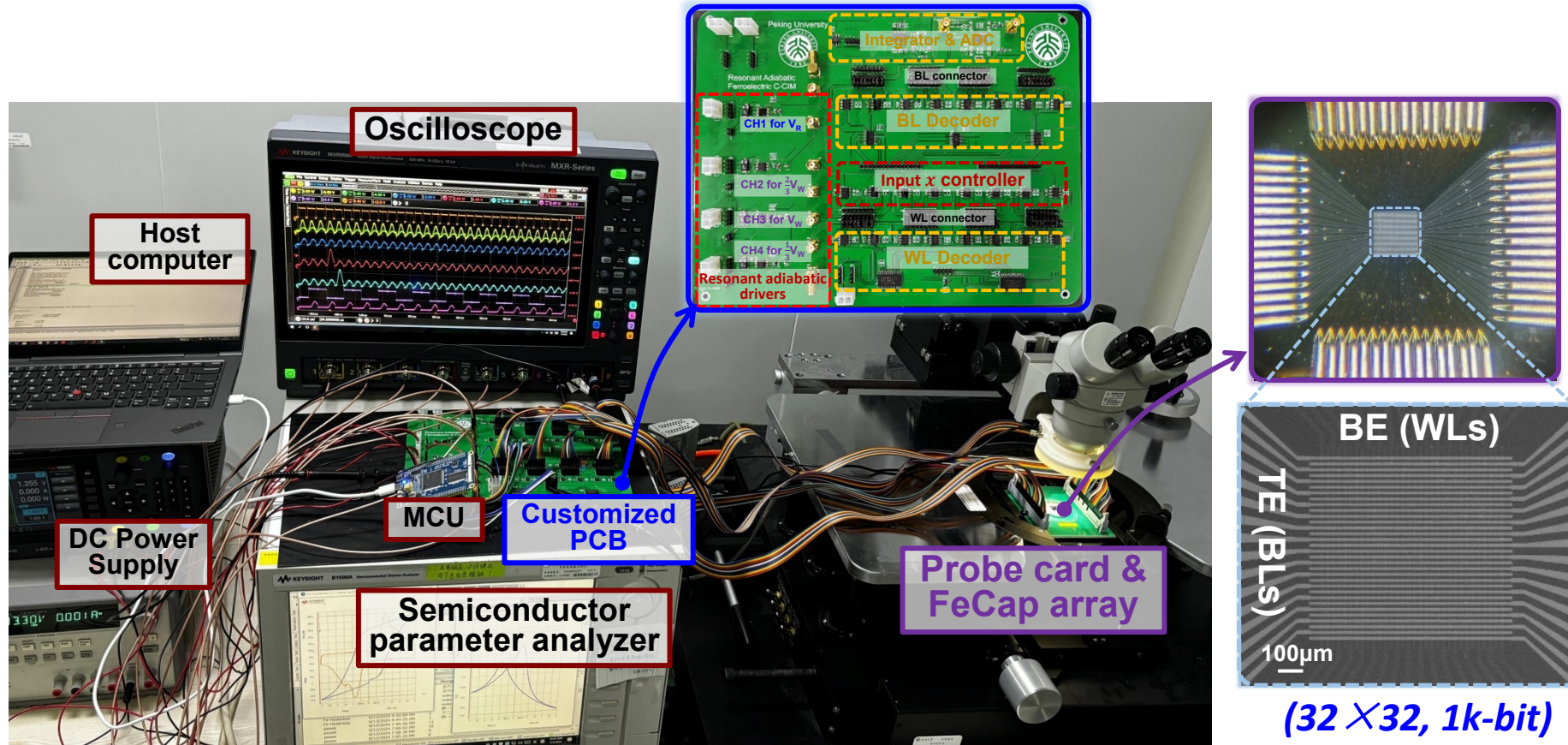
# Characteristics of FeCap array (2/2)

- Write operation with relatively high voltage ( $V_W$ )
  - $V_W \rightarrow$  Up (/down) polarization direction  $\rightarrow$  high (/low) capacitance state (HCS/LCS)
- Non-destructive readout operation with small  $V_R$ 
  - C-V curve shows negligible hysteresis for both HCS and LCS



The write and read operations of FeCap exhibit capacitive characteristic, w/o a DC path.

# Experimental setup of FeCap array-based resonant adiabatic (RA) C-CIM

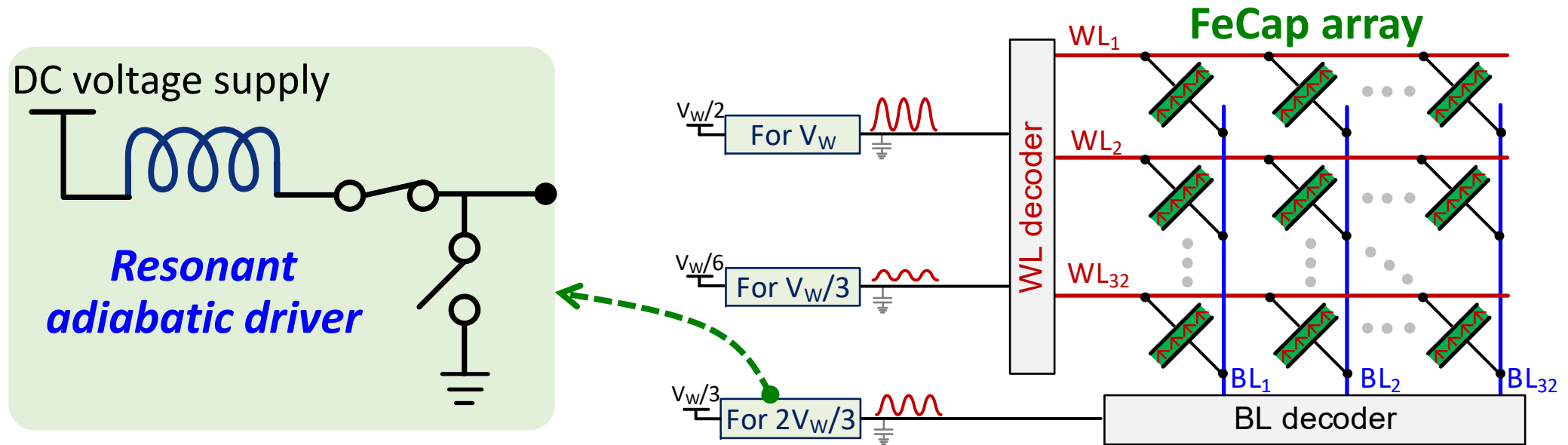


# Outline

- Design of FeCap array-based resonant adiabatic (RA) C-CIM
- Resonant adiabatic writing and computing operation
  - RA weight writing of FeCap array
  - RA MAC computing of FeCap array
- FeCap-based RA C-CIM macro for edge application
- Summary

# Resonant adiabatic writing operation (1/6)

- **$V_W/3$  program scheme** is used for the cross-point FeCap array
  - The FeCaps are initially set to an upward polarization state
  - $V_W$ ,  $V_W/3$ ,  $2V_W/3$  are generated by resonant adiabatic driver, and applied to WLs/BLs



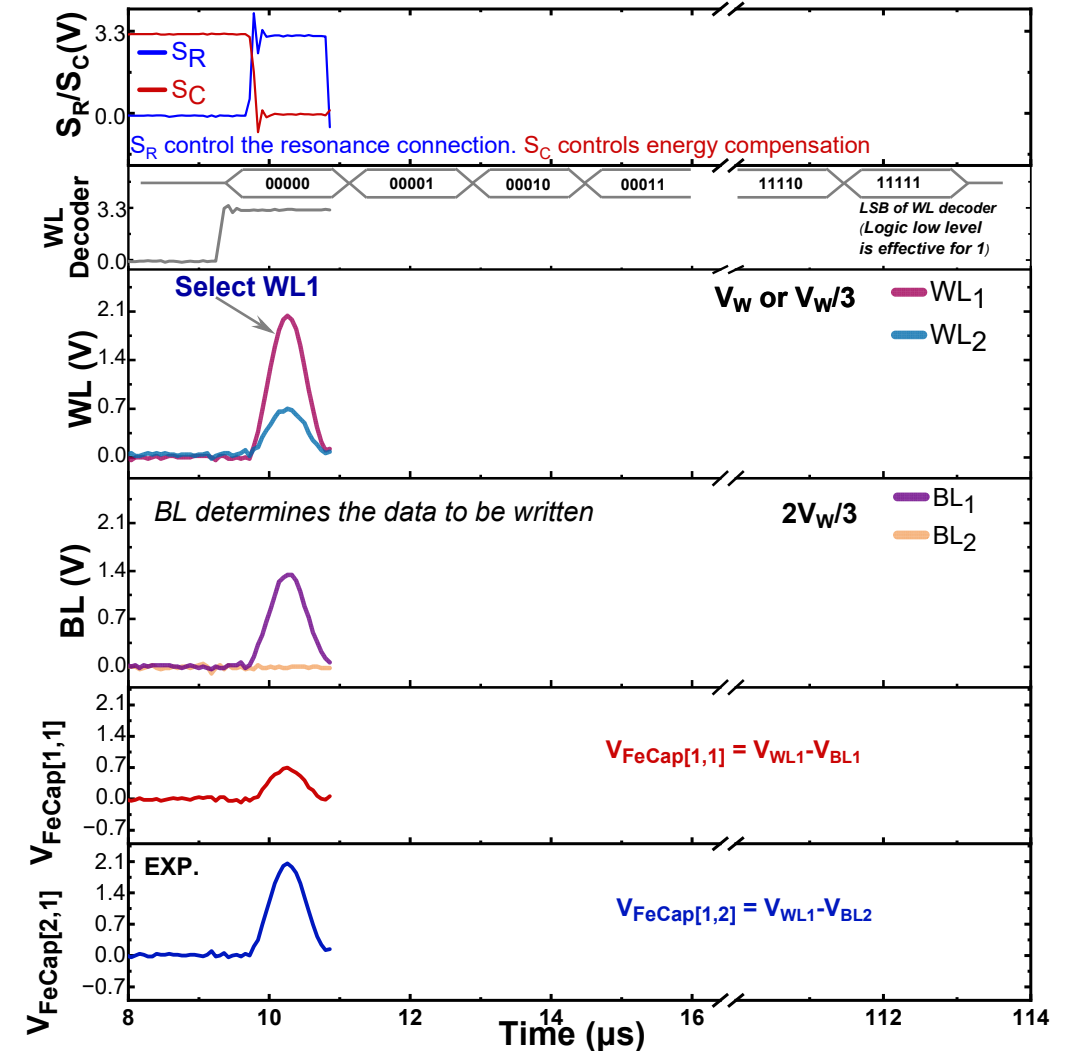
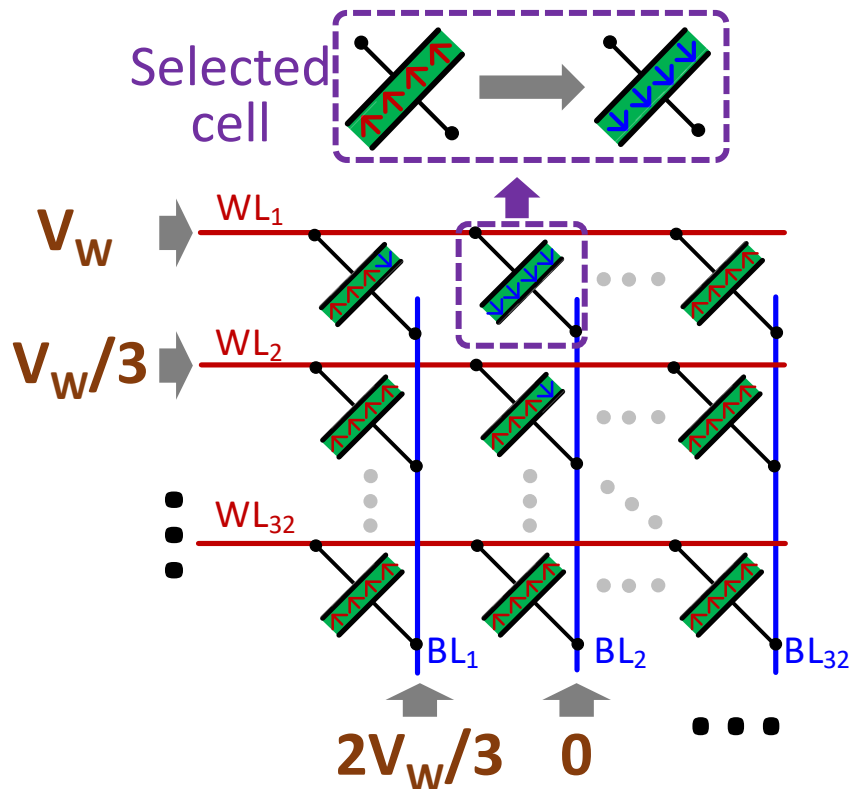
$V_W$  is applied to selected row.

$V_W/3$  is applied to unselected rows.

$2V_W/3$  is applied to bit-lines corresponding to the weight matrix value.

# Resonant adiabatic writing operation (2/6)

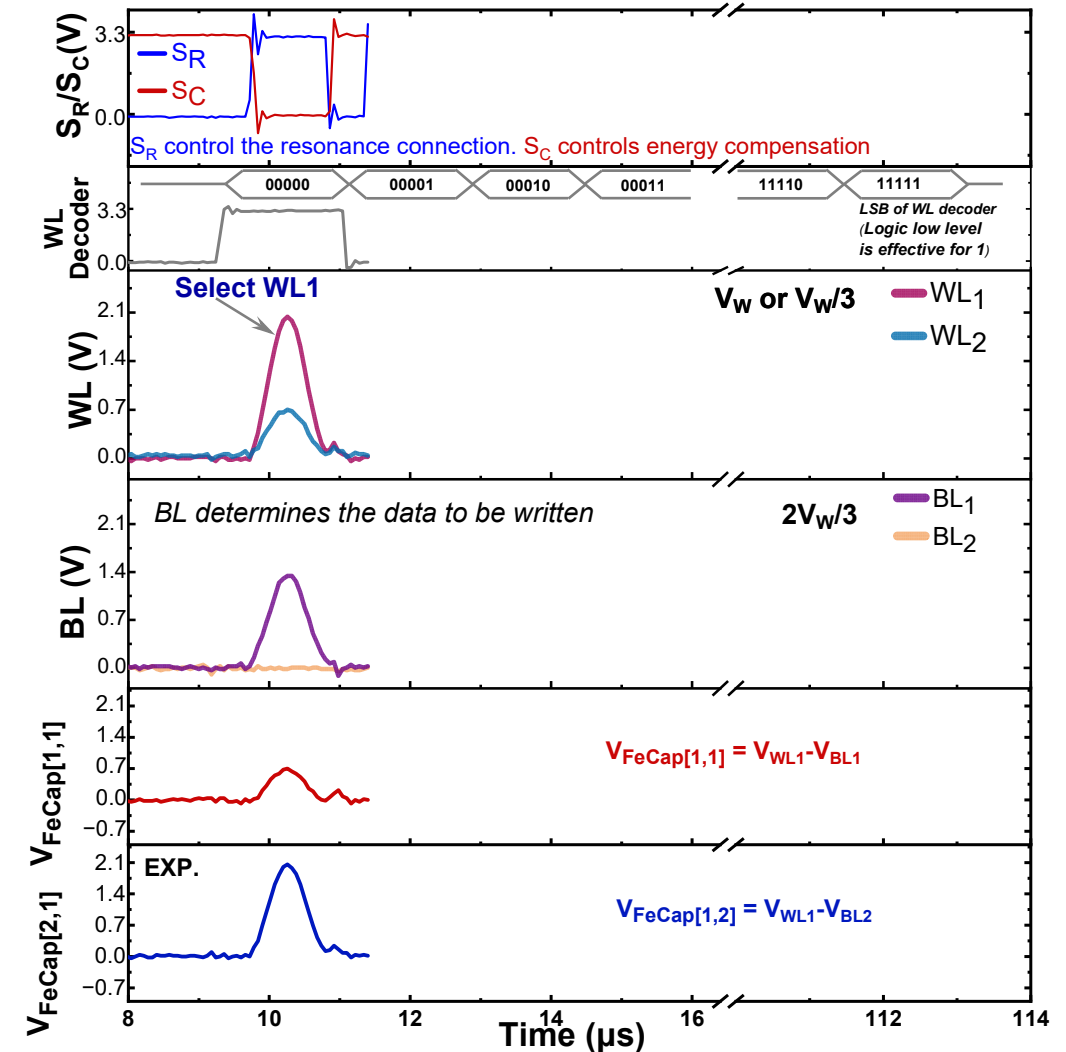
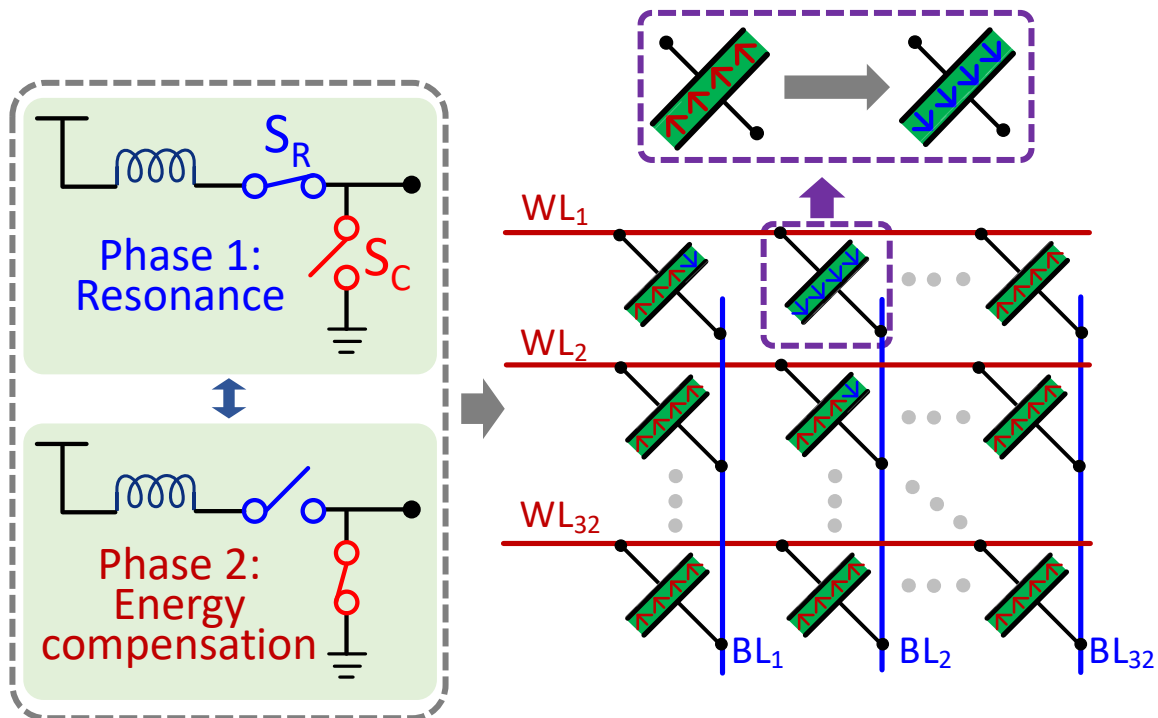
- **RA write cycle** for each row
  - Phase 1: write signal for the 1<sup>st</sup> row





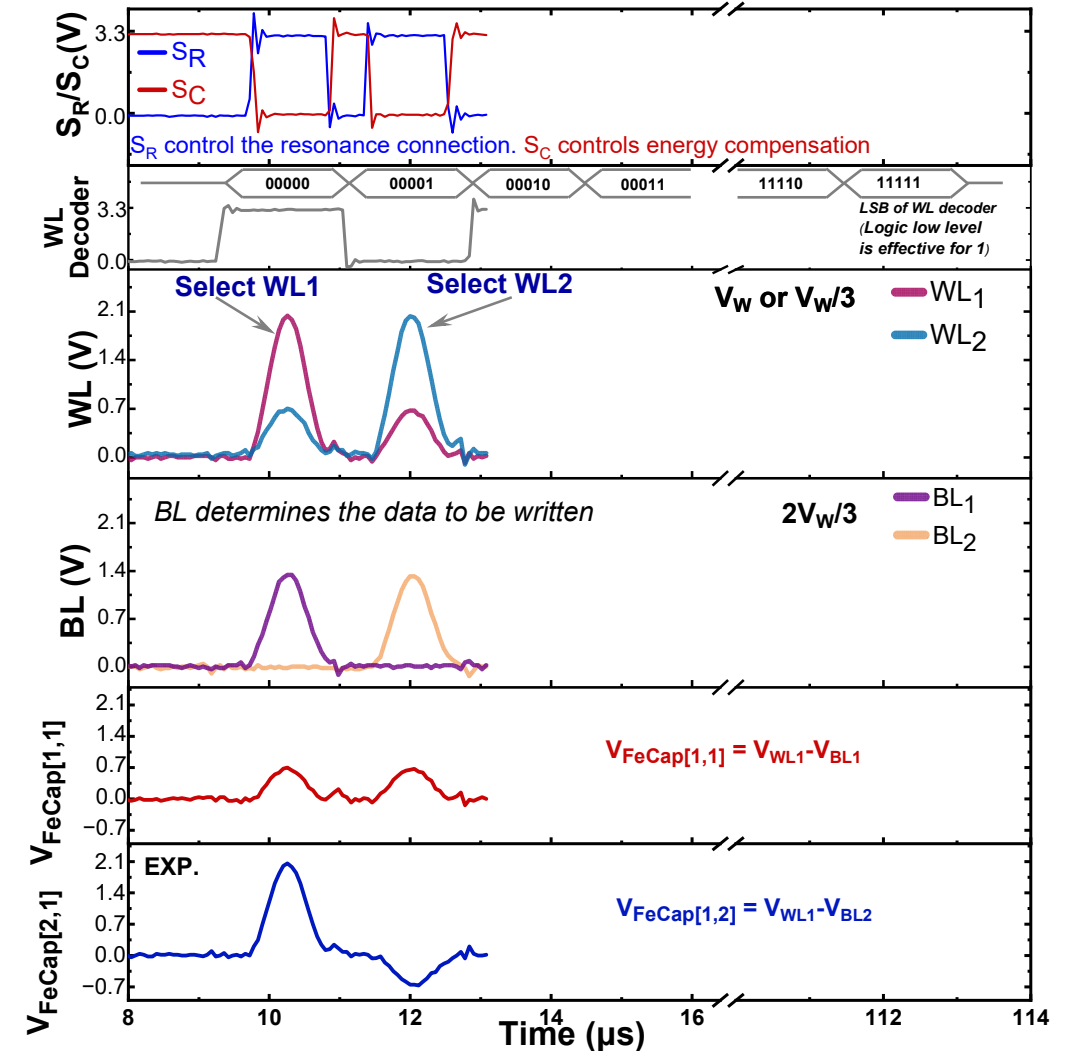
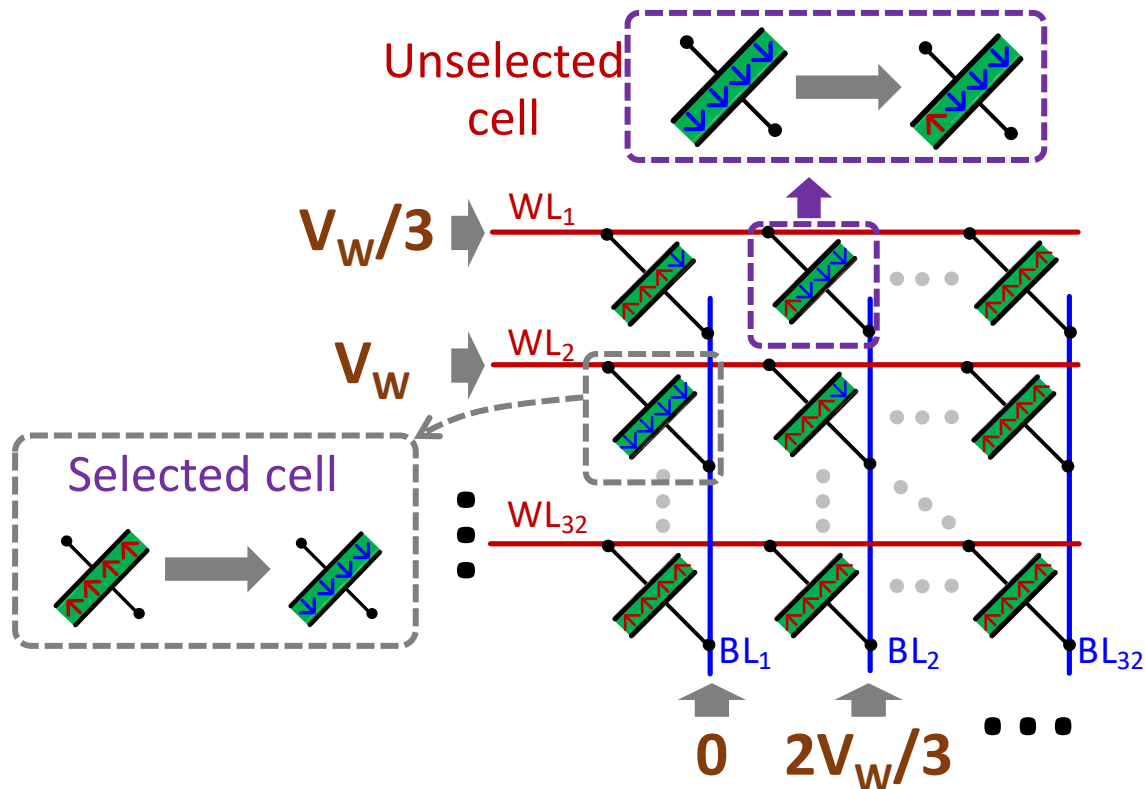
# Resonant adiabatic writing operation (3/6)

- **RA write cycle** for each row
  - Phase 1: write signal for the 1<sup>st</sup> row
  - Phase 2: energy compensation



# Resonant adiabatic writing operation (4/6)

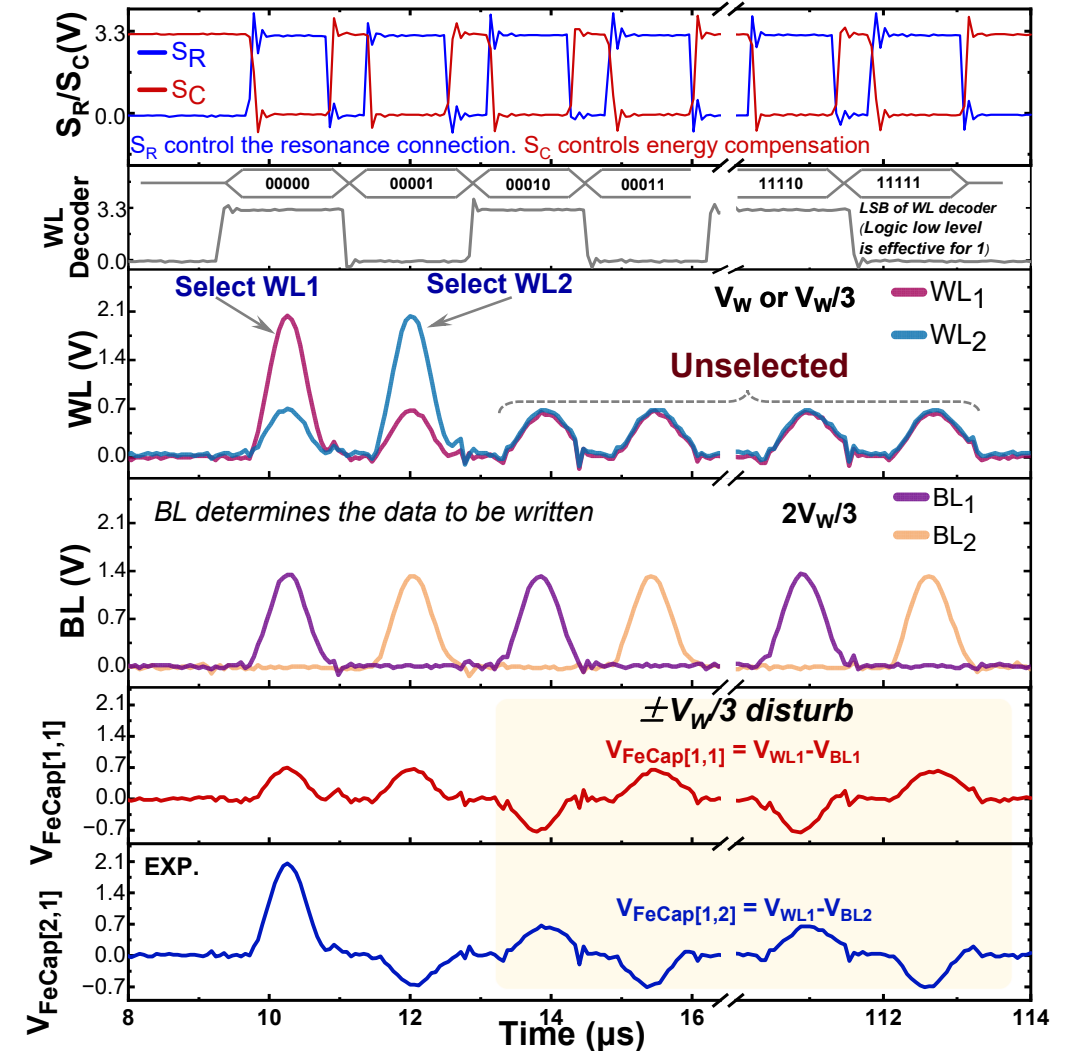
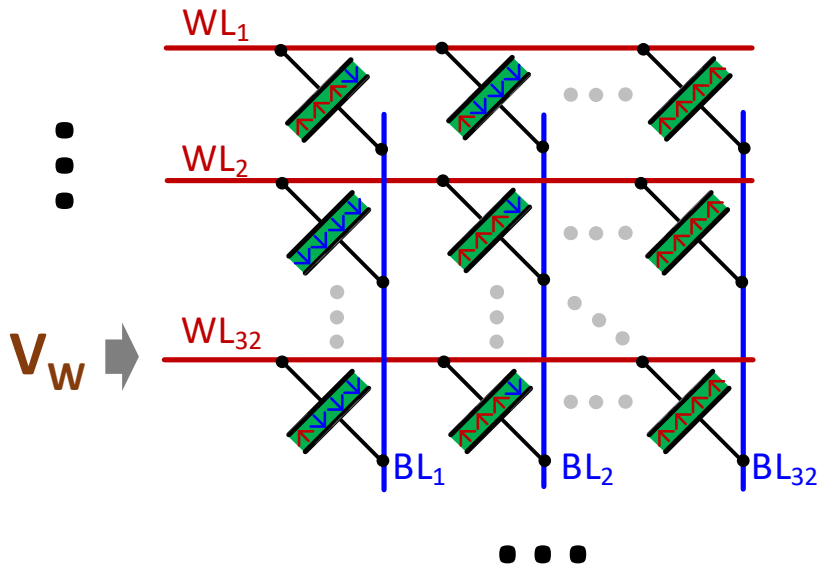
- Row-by-row writing operation
  - RA write cycle for 2<sup>nd</sup> row





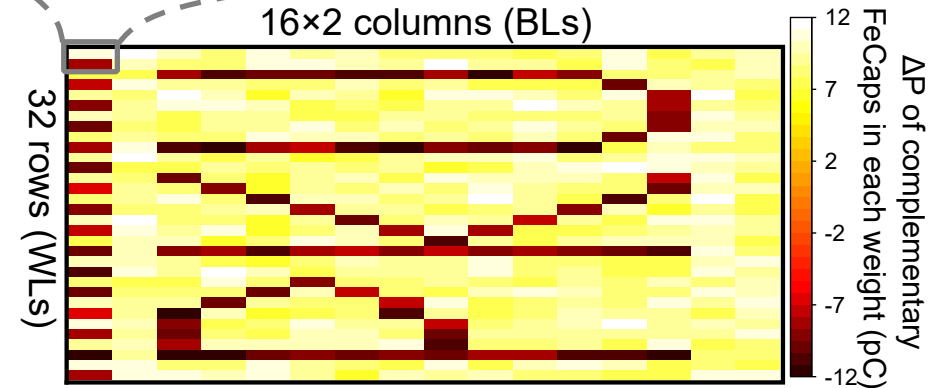
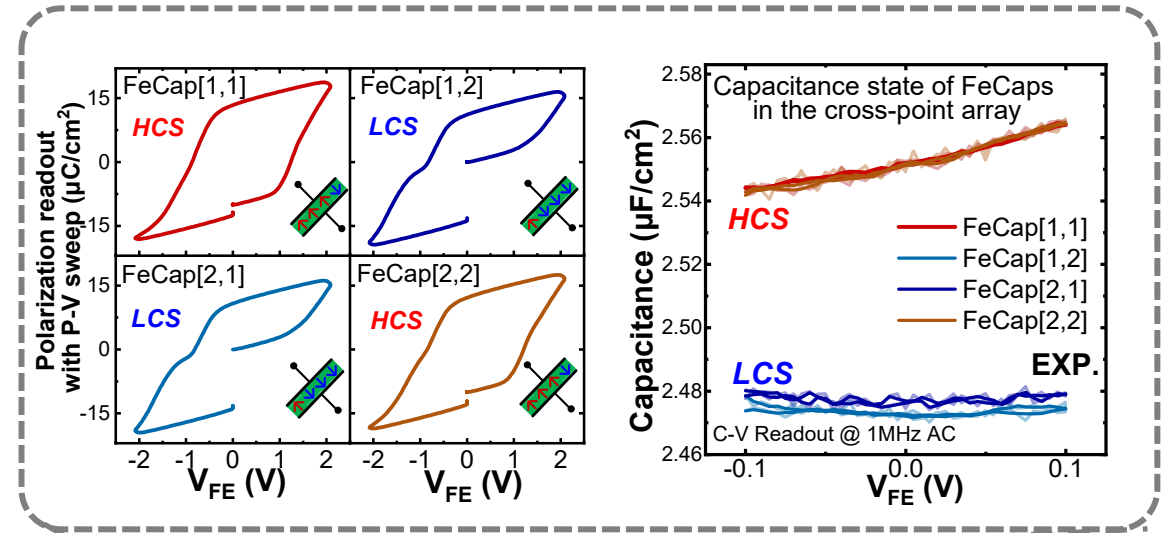
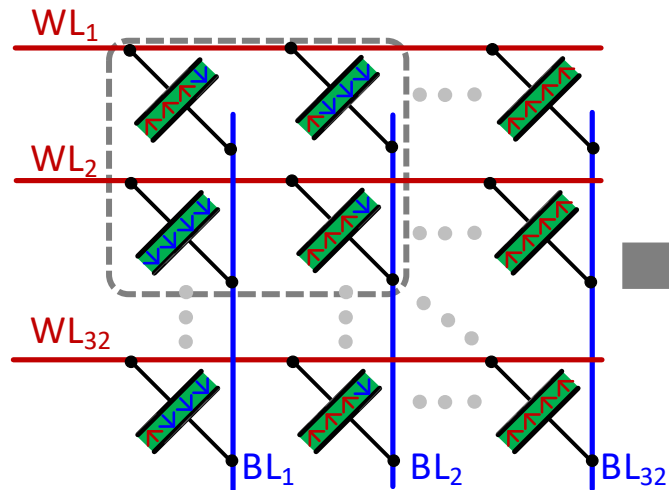
# Resonant adiabatic writing operation (5/6)

- Row-by-row writing operation
  - RA write cycle for 2<sup>nd</sup> row
  - ⋮
  - RA write cycle for 32<sup>nd</sup> row



# Resonant adiabatic writing operation (6/6)

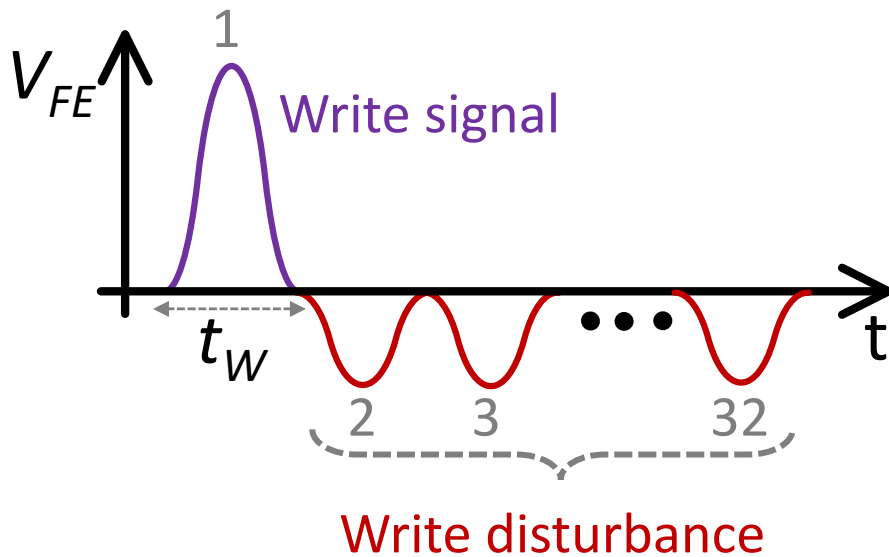
- Row-by-row writing operation
  - RA write cycle for 2<sup>nd</sup> row
  - RA write cycle for 32<sup>nd</sup> row



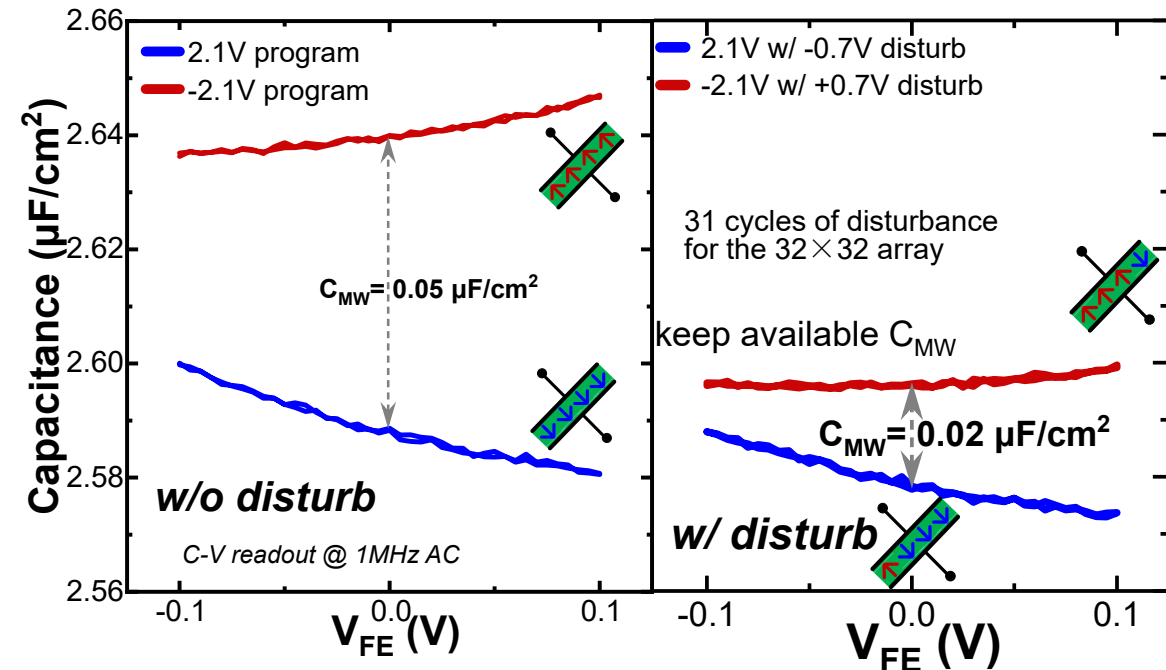
Mapping diagram of weight matrix

# Experimental analysis of write disturbance (1/2)

- Write disturbance of  $V_W/3$  scheme for the cross-point FeCap array
  - Remain the available capacitive memory window ( $C_{MW}$ ) under disturbance

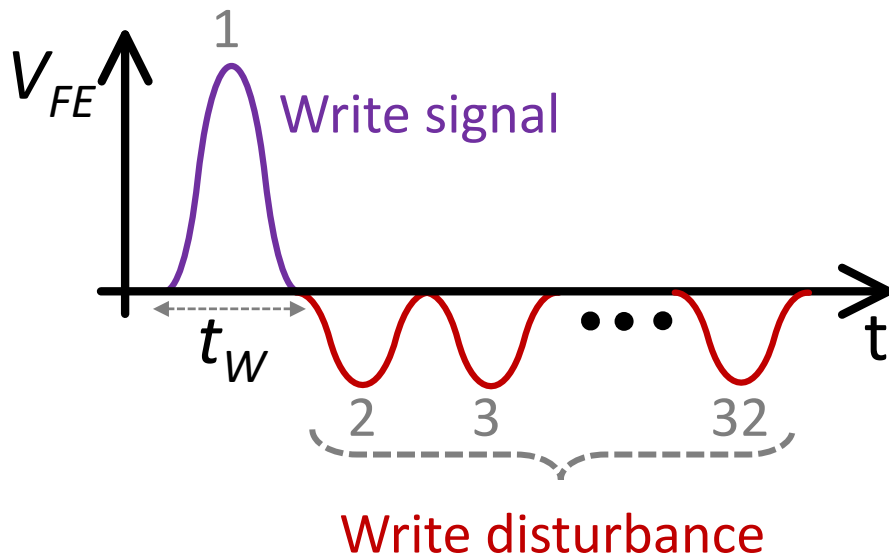


A case of weight write in an FeCap

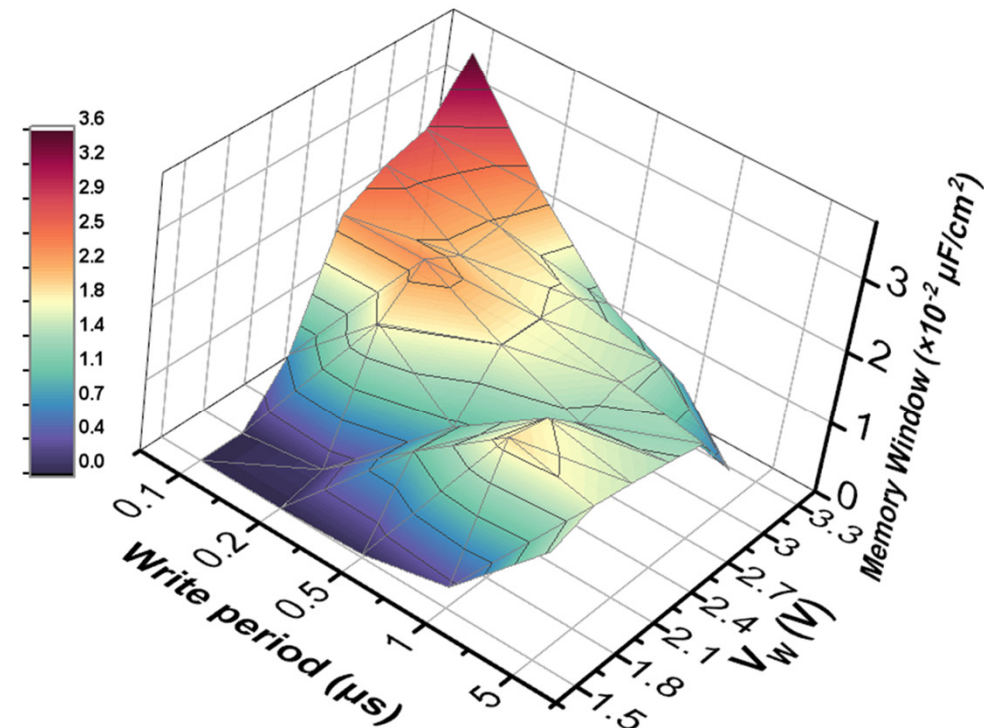


# Experimental analysis of write disturbance (2/2)

- $C_{MW}$  after disturbance is not completely monotonic with  $V_W$  and  $t_W$ 
  - The electric field and time dependent FE switching  $\rightarrow$  Increased  $V_W$  and write period ( $t_W$ ) cause more polarization switching, as well as more disturbance
  - Considering power consumption,  $V_W=2.1V$  and  $t_W=1\mu s$  are used for RA writing

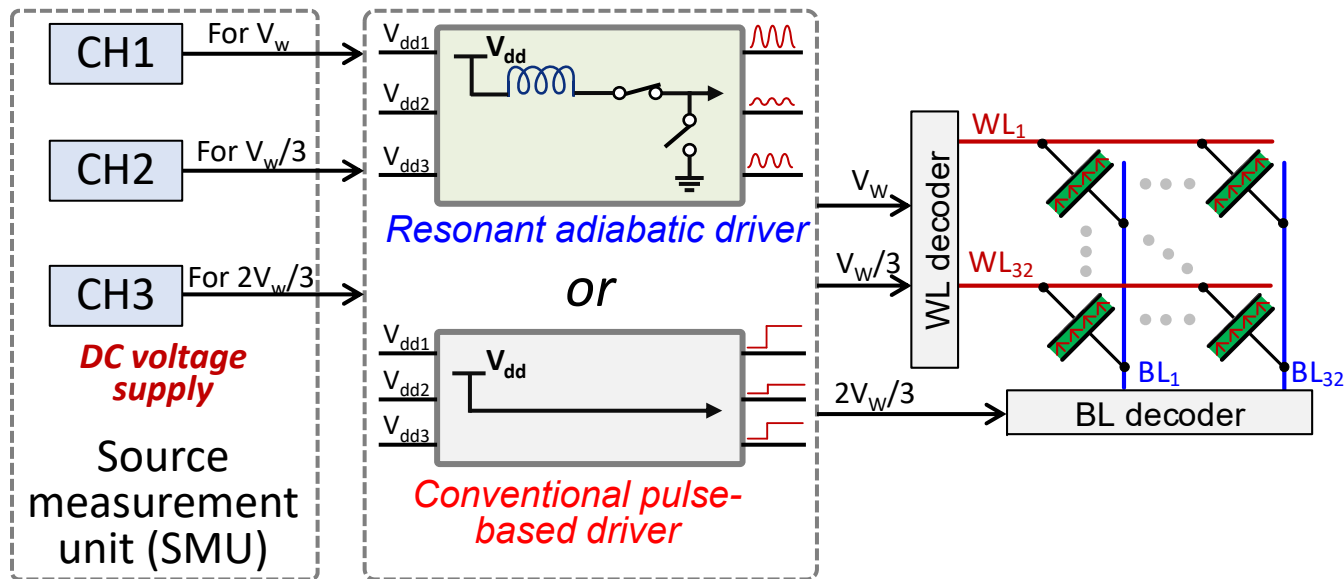


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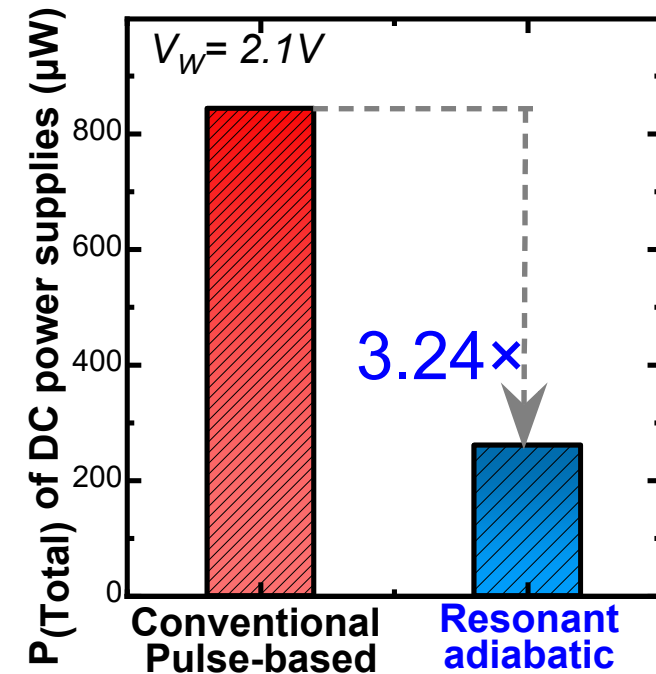


# Experimental analysis of write power

- Measured power consumption during writing operation
  - The resonant adiabatic scheme achieves significant **reduction in total write power consumption due to charge recycling**



Measurement setup

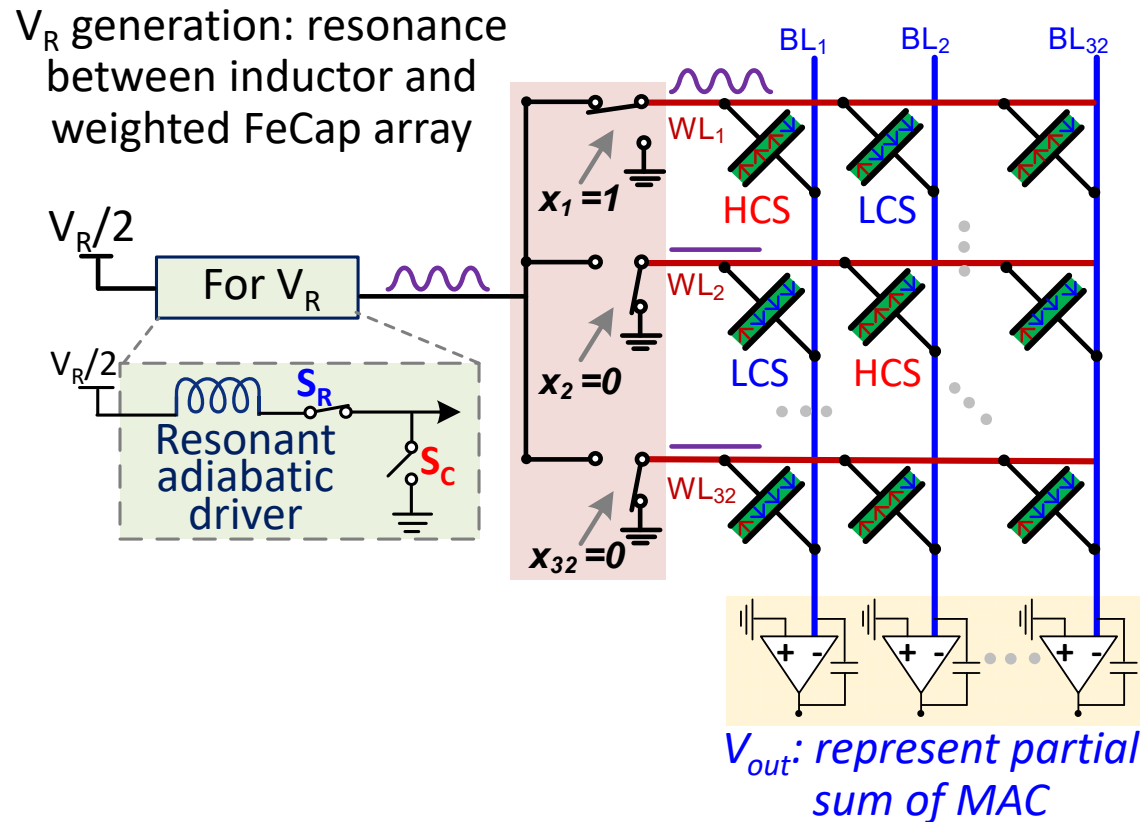


# Outline

- Design of FeCap array-based resonant adiabatic (RA) C-CIM
- Resonant adiabatic writing and computing operation
  - RA weight writing of FeCap array
  - RA MAC computing of FeCap array
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# Resonant adiabatic architecture for MAC (1/2)

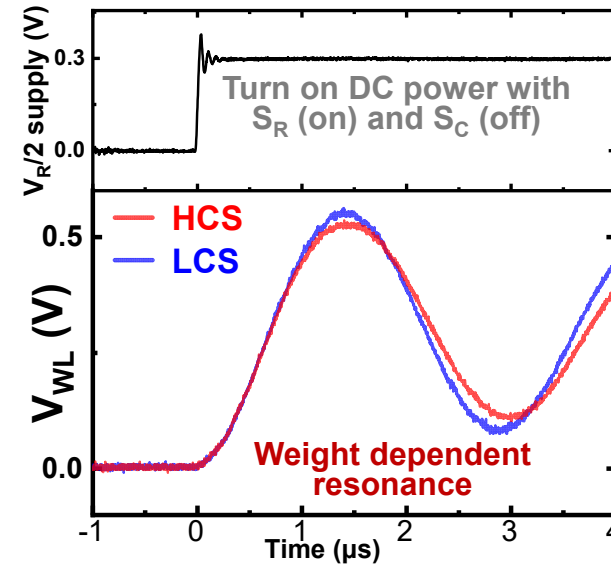
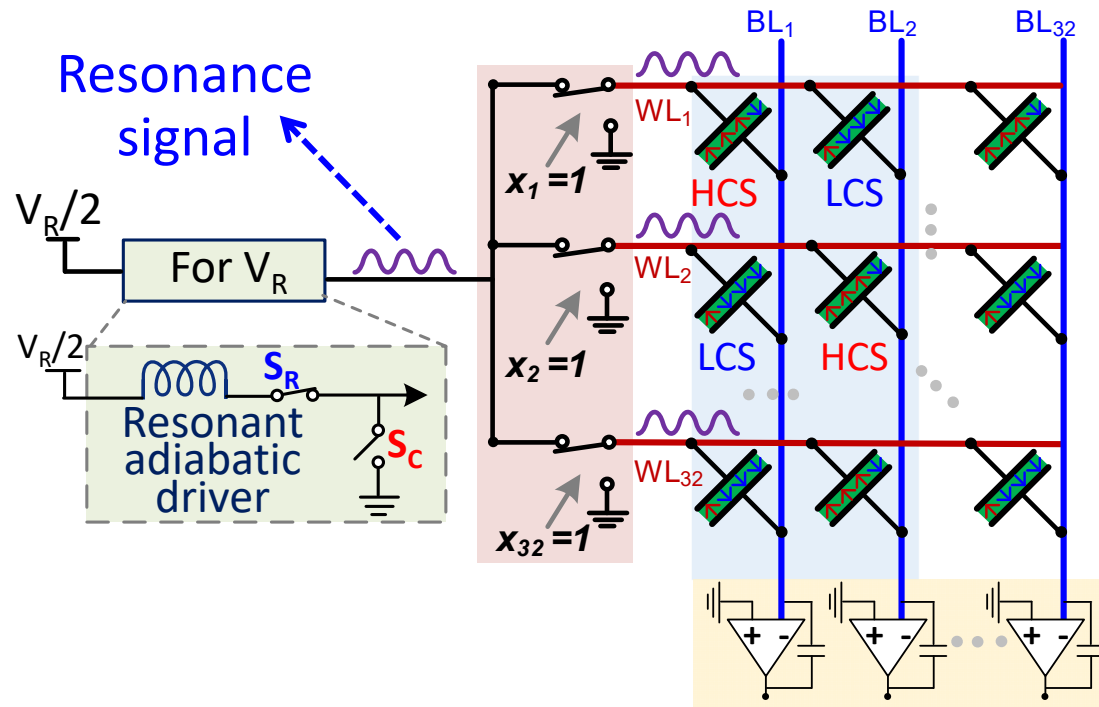
- Circuit architecture for resonant adiabatic (RA) MAC computing



- **Weight ( $W$ ) matrix:** capacitance states of FeCap array
- **Input ( $X$ ) vector:** row-wise switches  
→ connect WLs to  $V_R$  (if  $x=1$ ) or GND (if  $x=0$ )
- **Partial sum ( $V_{out}$ ):** induced charge of each column is summed in charge domain through the integrators

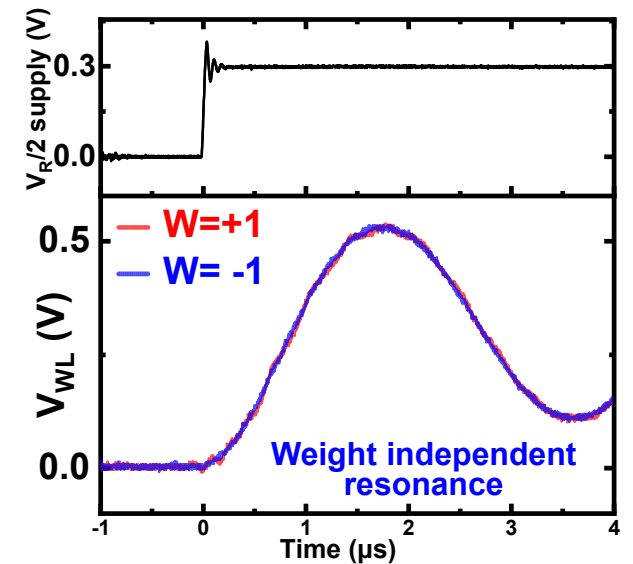
# Resonant adiabatic architecture for MAC (2/2)

- **Complementary weight storage** in FeCap array
  - Total load capacitance is independent with weight data → Stabilize resonant signal ( $V_R$ )



Case1: Single-end storage

Unstable resonance  
for various weights



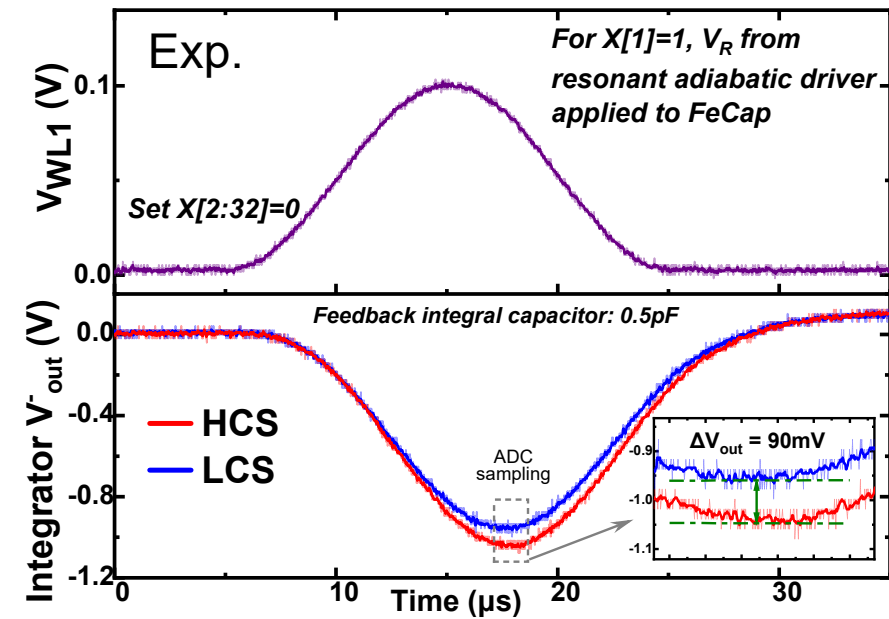
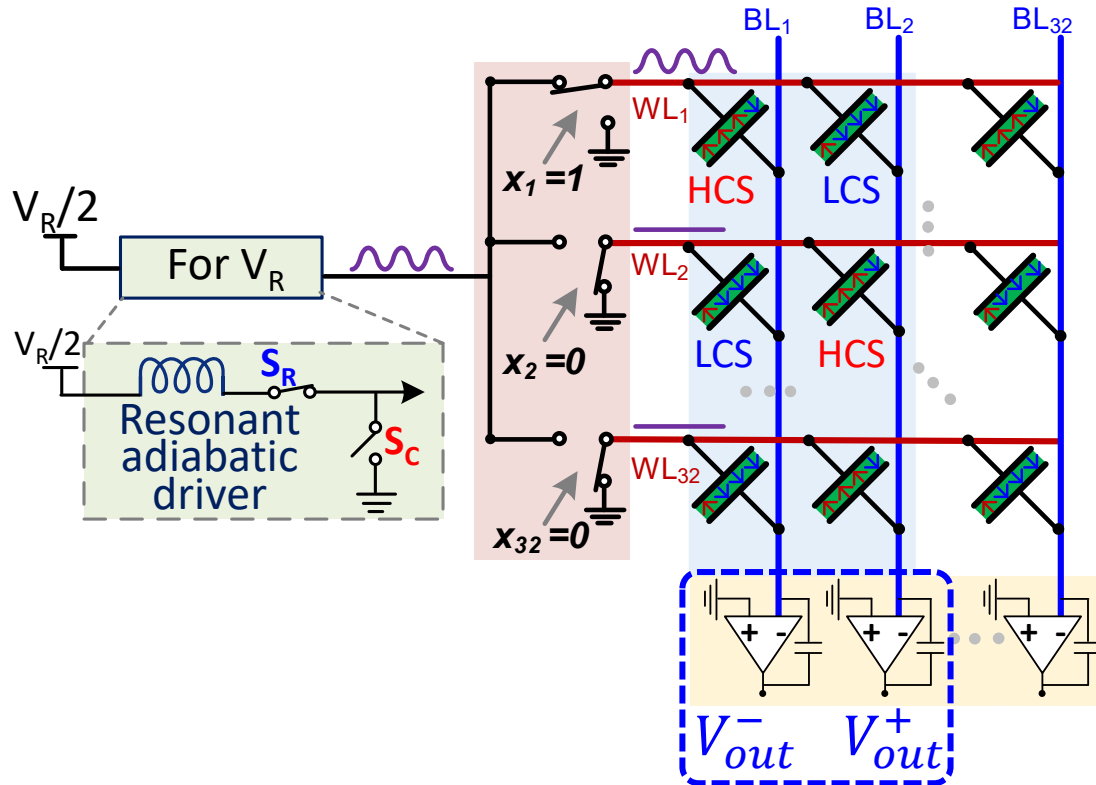
Case2: Complementary storage

Stable resonance  
for various weights



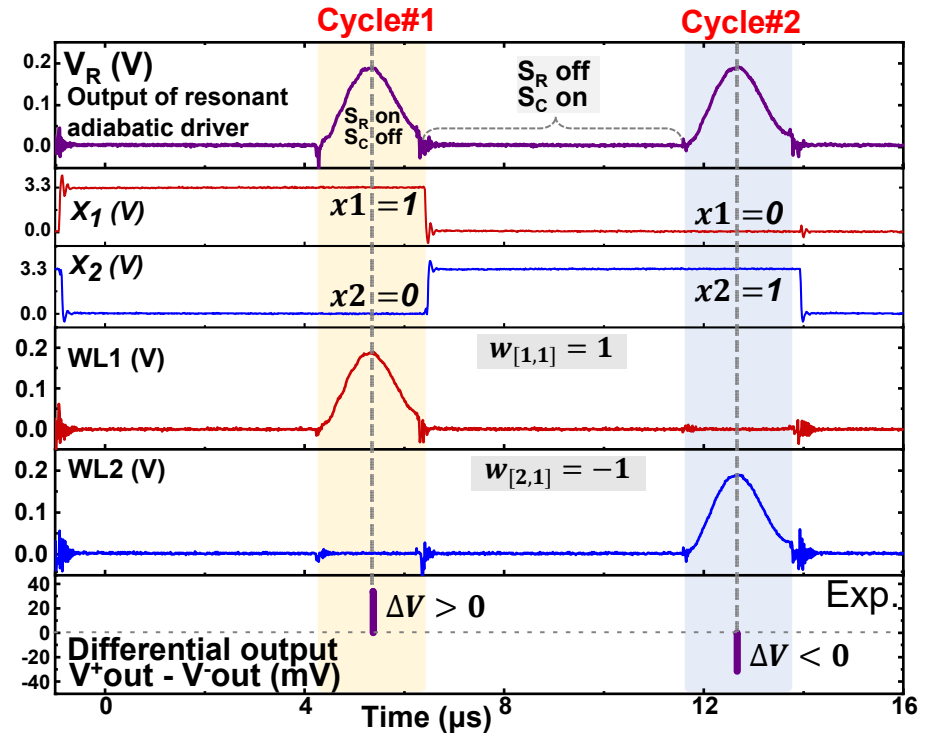
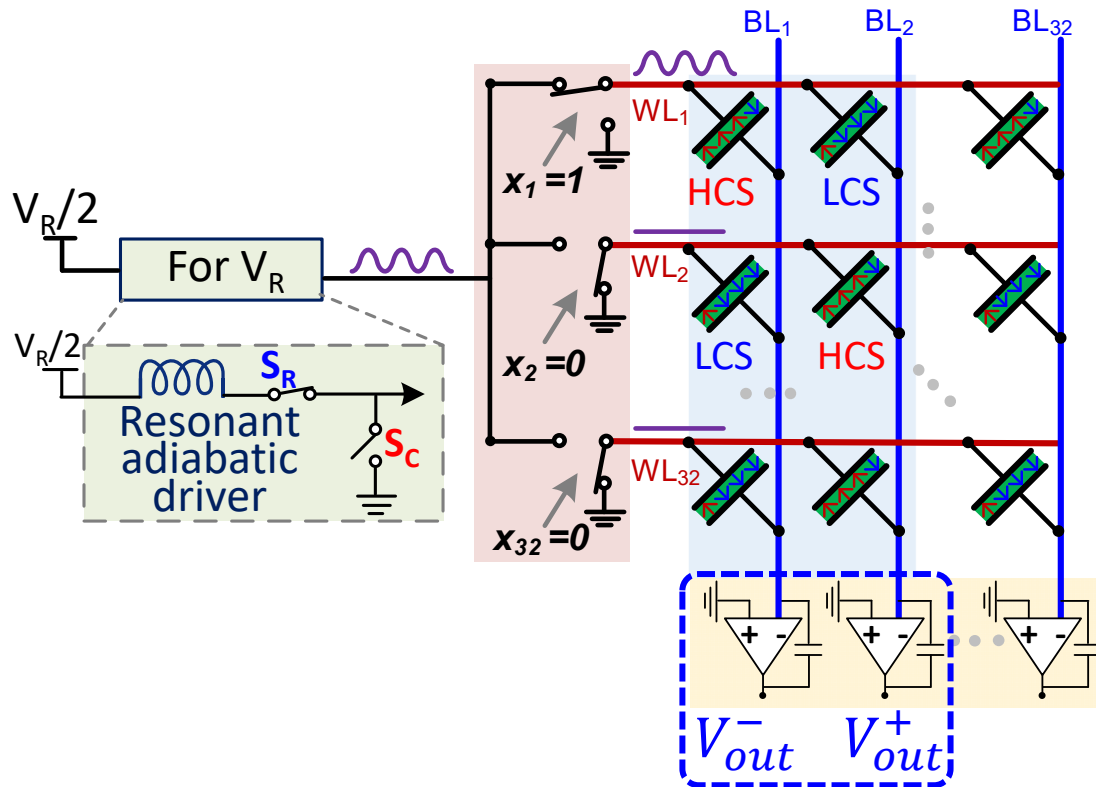
# Resonant adiabatic computing (1/3)

- Resonant adiabatic based multiplication operation
  - For FeCap in HCS/LCS, the input  $V_R$  causes  $V_{out}$  with high/low amplitude
  - The differential voltage, corresponding to complementary weight, represents  $x_i \cdot w_{ij}$



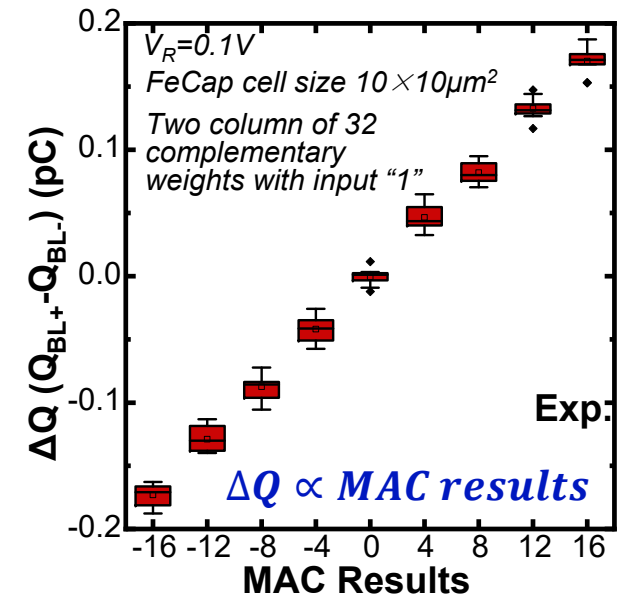
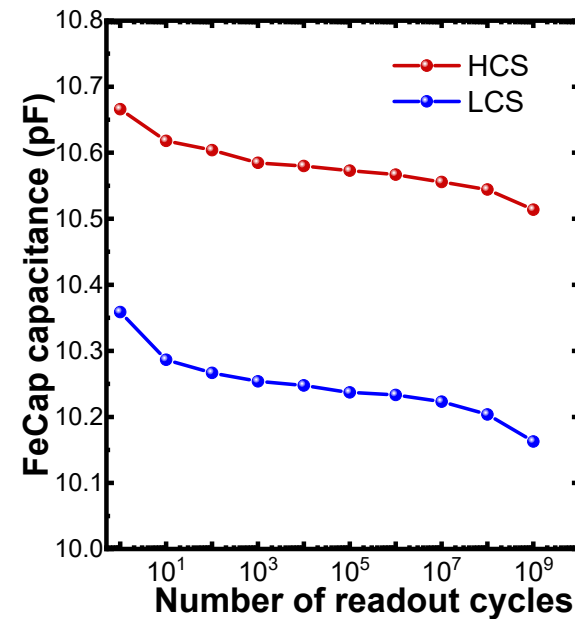
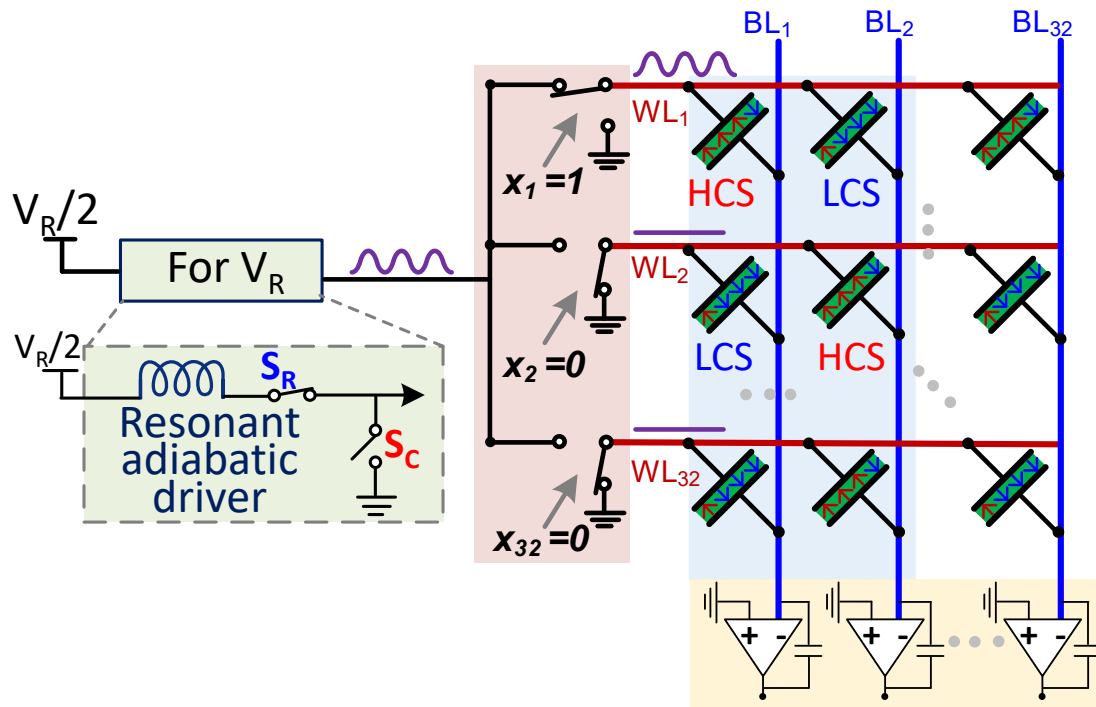
# Resonant adiabatic computing (2/3)

- Charge-domain resonant adiabatic multiply-accumulate (MAC)
  - The induced charge is naturally shared across the column



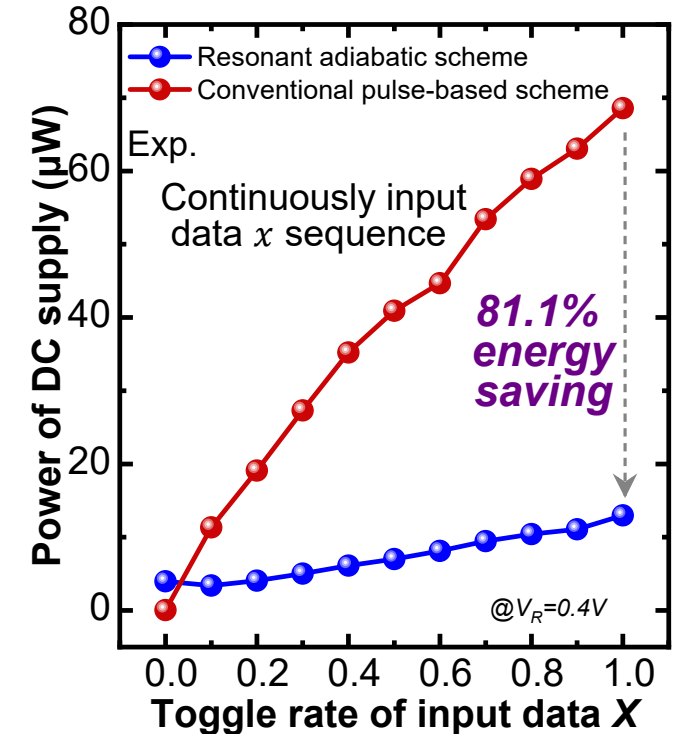
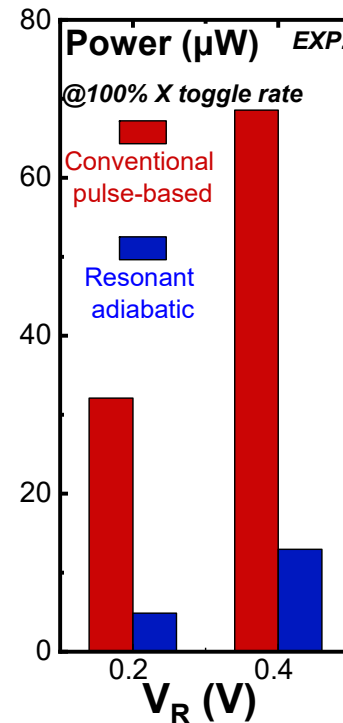
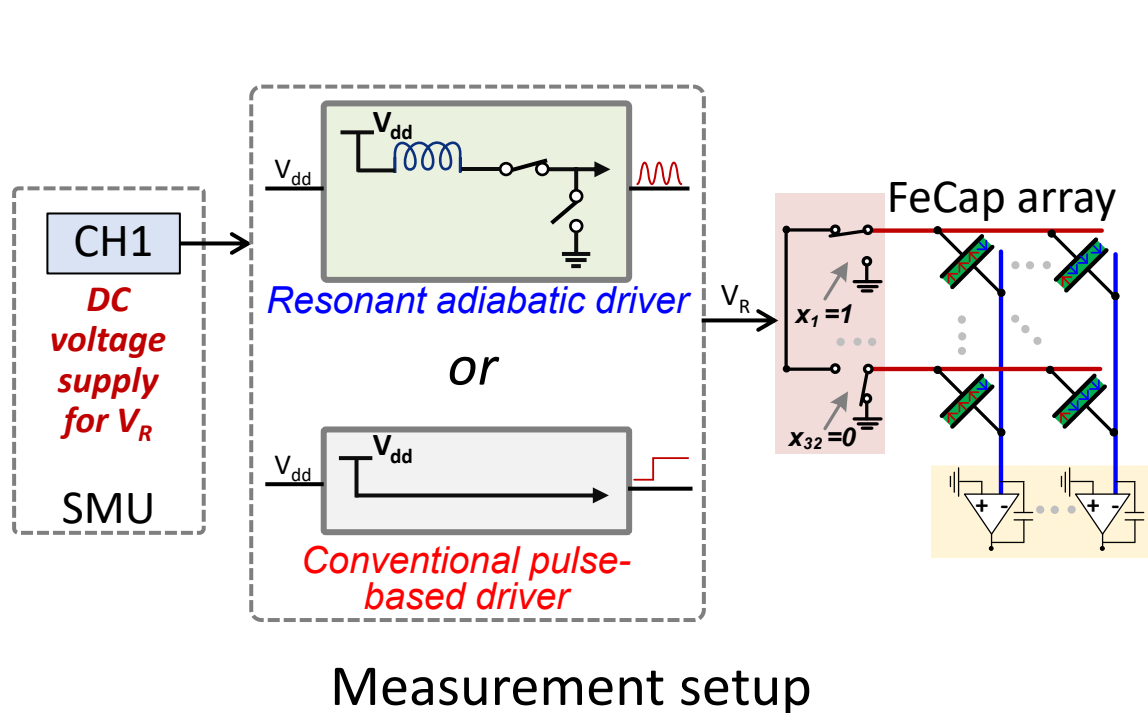
# Resonant adiabatic computing (3/3)

- The  $C_{MW}$  of complementary weight remains stable during cycling computing
- The cumulative charge exhibits a linear relationship with the MAC results



# Experimental analysis of power consumption

- Resonant adiabatic C-CIM significantly improves energy efficiency for MAC
  - Most of the energy is saved in resonant adiabatic scheme due to charge recycling

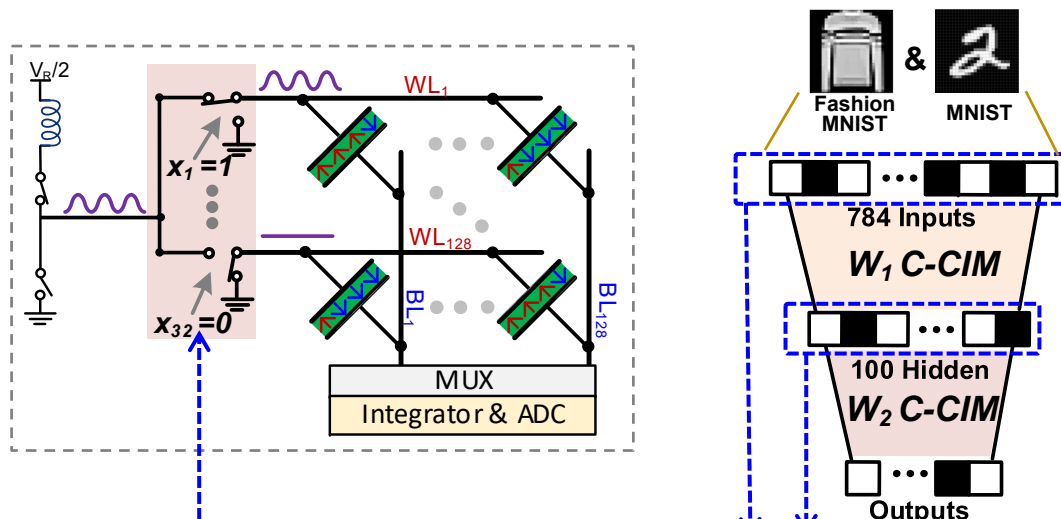


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- Summary

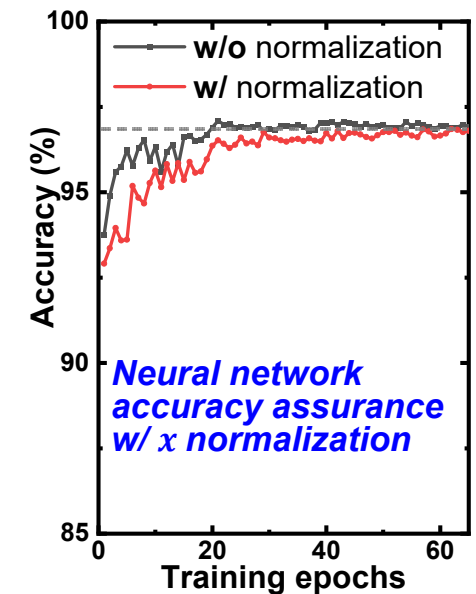
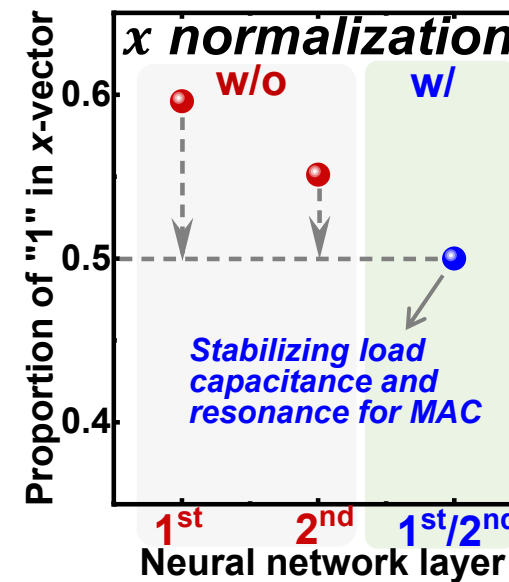
# Pattern recognition with FeCap-based RA C-CIM

- ANN structure based on proposed FeCap-based resonant adiabatic C-CIM
  - **Co-design of algorithm and RA C-CIM circuit:** layer normalization algorithm  $\rightarrow$  symmetrical distribution of activations ( $x$ )  $\rightarrow$  constant load capacitance
  - High accuracy for pattern recognition



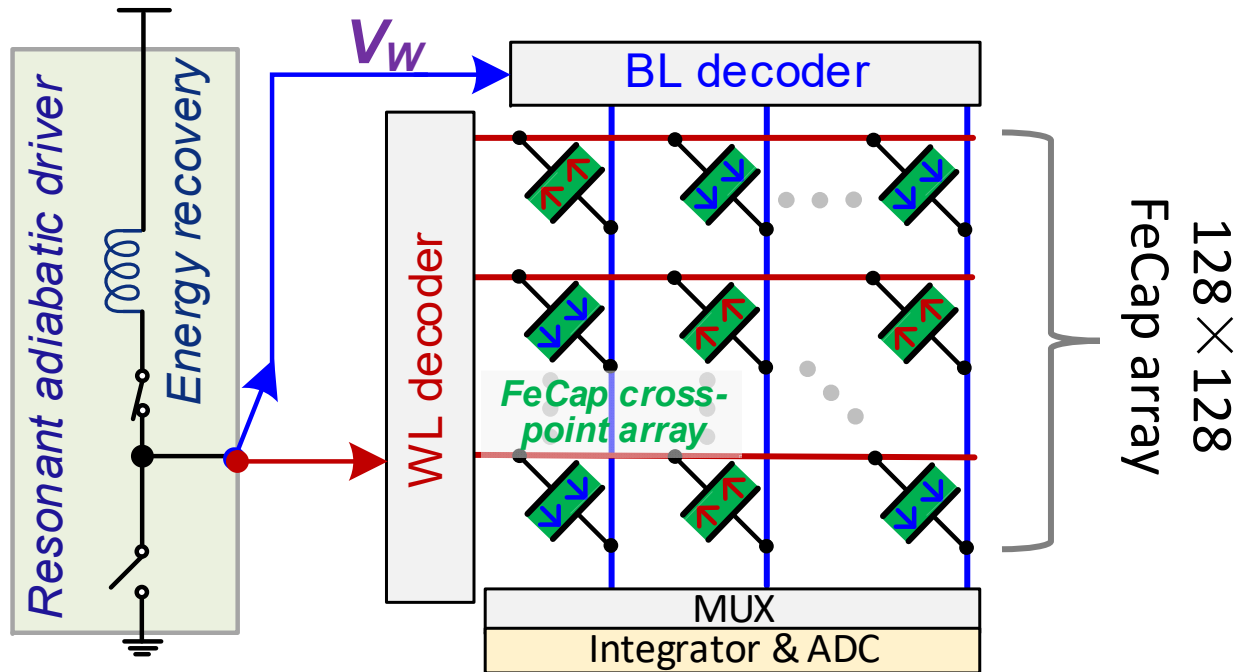
$$x_i = \text{sign}(y_i - \text{MEDIAN}\{Y\})$$

$y$  is the weighted sum of last layer

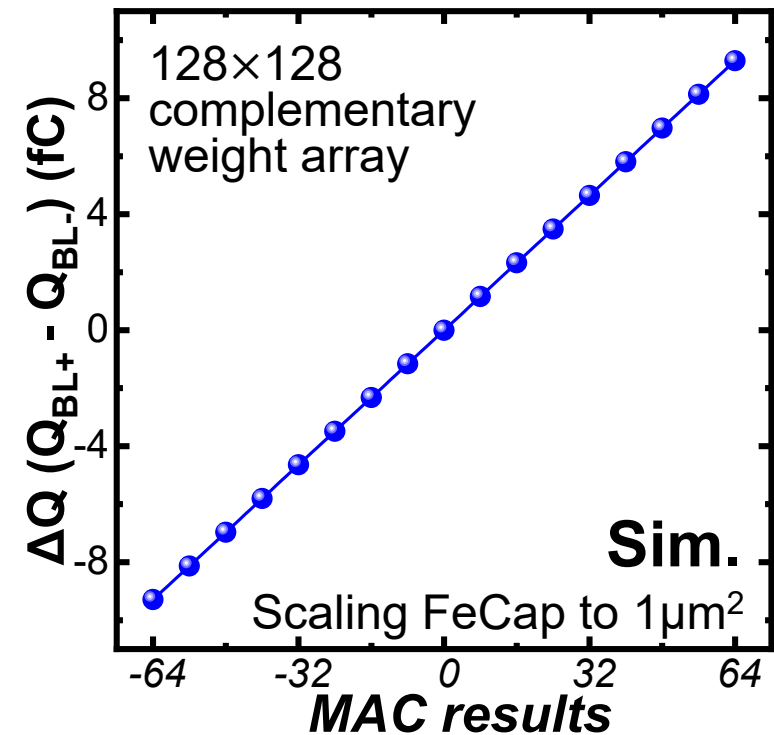


# Performance evaluation of FeCap-based RA C-CIM (1/2)

- FeCap-based resonant adiabatic C-CIM array
  - Implement charge domain MAC computing with good linearity

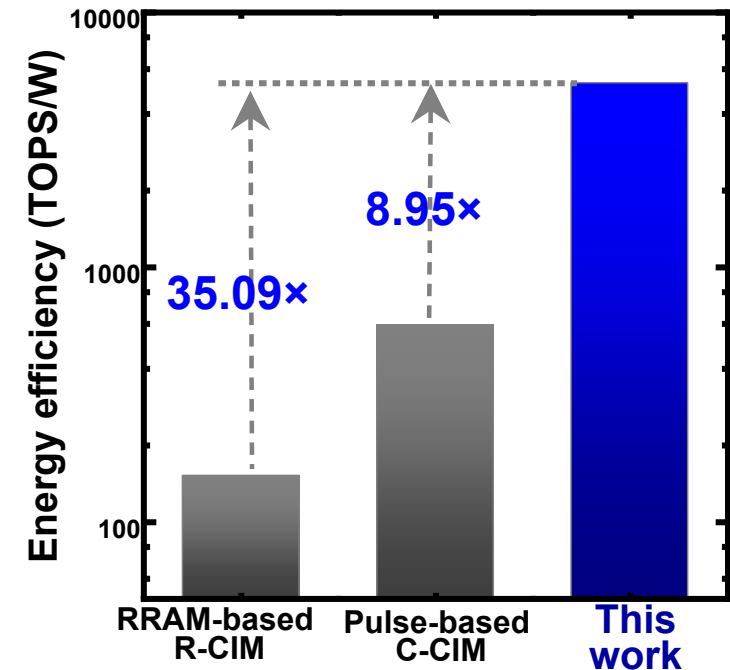
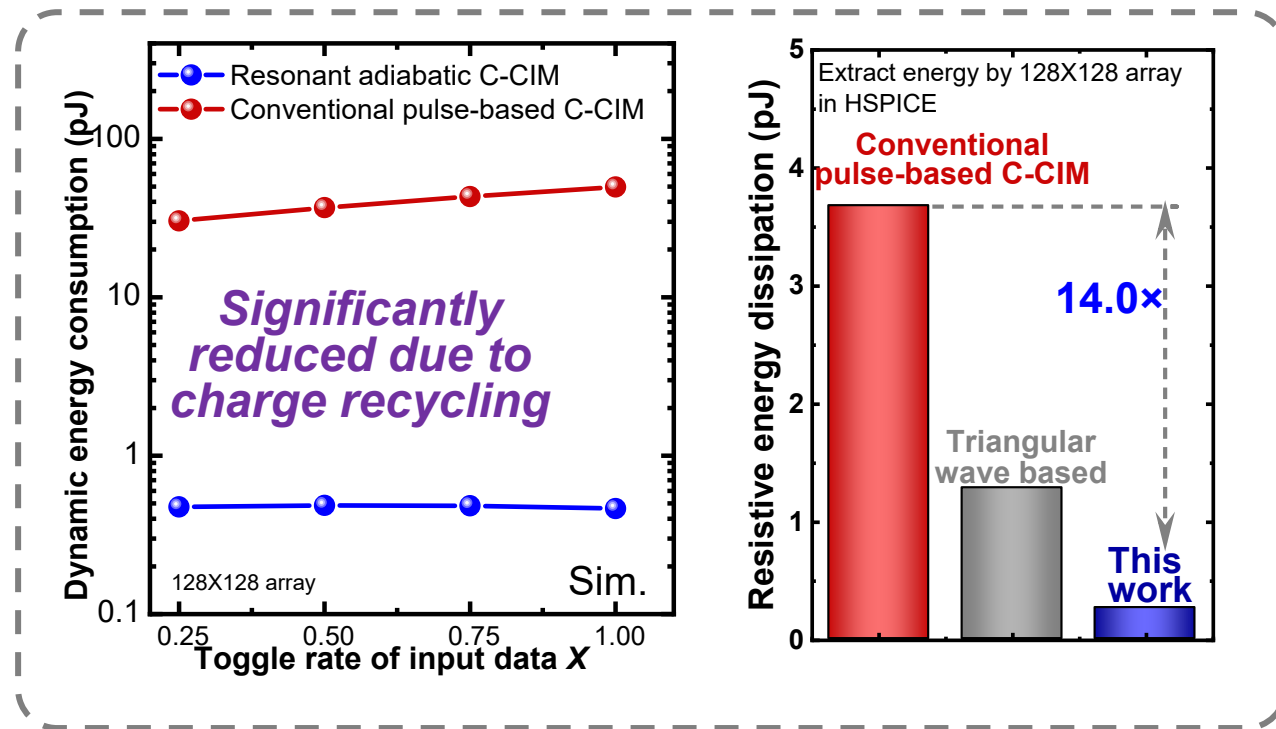


FeCap-based RA C-CIM system in HSPICE simulation  
(FeCap model based on experimentally extracted capacitance characteristics)



# Performance evaluation of FeCap-based RA C-CIM (2/2)

- Energy efficiency of proposed FeCap-based resonant adiabatic C-CIM
  - Dynamic power and resistive dissipation are reduced due to resonant adiabatic design
  - Improved energy efficiency compared with conventional R-CIM and pulse-based C-CIM





# Benchmark of the proposed FeCap-based resonant adiabatic C-CIM with other reported solutions.

|                             | [1]                    | [2]                       | [3]                                    | [5-6]                    | <b><i>This work</i></b>                |
|-----------------------------|------------------------|---------------------------|----------------------------------------|--------------------------|----------------------------------------|
| Memory type                 | RRAM                   | FeRAM (PZT)               | Charge-trapping MOS Cap                | FeCap                    | <b><i>FeCap (HZO)</i></b>              |
| Storage & Readout principle | Resistance             | FE polarization switching | Capacitance memory window ( $C_{MW}$ ) |                          | <b><math>C_{MW}</math></b>             |
| CIM type                    | R-CIM                  | /                         | C-CIM                                  | C-CIM                    | <b><i>C-CIM</i></b>                    |
| Weight write scheme         | Current                | Adiabatic                 | Pulse-based                            | Conventional pulse-based | <b><i>Resonant adiabatic</i></b>       |
| MAC scheme                  | Current                | /                         | Adiabatic (Sim)                        |                          |                                        |
| Writing energy (pJ/bit)     | 4.5                    | 0.99                      | high (>5V, >1ms)                       | 0.82 <sup>(a)</sup>      | <b><i>0.25pJ/bit<sup>(a)</sup></i></b> |
| Subarray energy             | 210.7pJ <sup>(b)</sup> | /                         | 95.6pJ                                 | 49.7pJ <sup>(b)</sup>    | <b><i>0.46pJ</i></b>                   |
| Energy efficiency (TOPS/W)  | 151.4 <sup>(bc)</sup>  | /                         | 3452.6                                 | 591.6 <sup>(c)</sup>     | <b><i>5298 TOPS/W</i></b>              |

(a) Extracted from experimental measurement. (b) Calculated in HSPICE based on same array size with respectively proposed weight cell characteristics. (c) The energy of ADC is extracted from [6].

[1] P. Yao, et al., *Nature*, 2020; [2] S. Jeloka, et al., *VLSI*, 2018; [3] K. U. Demasius, et al., *Nat. Electron*, 2021; [4] Y. C. Luo, et al., *TCAS-II*, 2021; [5] S. Mukherjee, et al., *IEDM*, 2023; [6] L. Xu, et al., *DAC*, 2022

# Outline

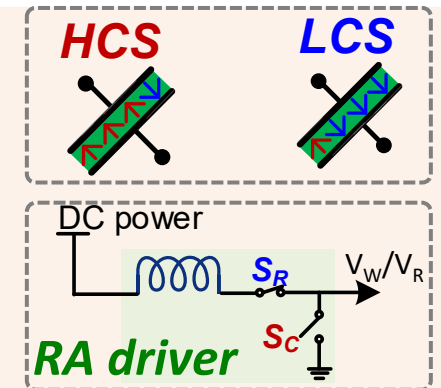
- Design of FeCap array-based resonant adiabatic (RA) C-CIM
- Resonant adiabatic writing and computing operation
- FeCap-based RA C-CIM macro for edge application
- Summary

# Summary

A novel **resonant adiabatic (RA) capacitive CIM (C-CIM)** based on **FeCap array**, enabling **RA writing and computing** with high energy-efficiency

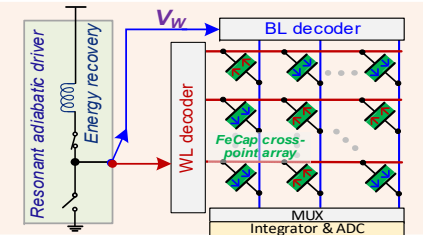
## **Co-design of ferroelectric device & adiabatic circuit technique**

- FeCap: non-volatile ferroelectric (FE) polarization modulated capacitance & **capacitive property of both write and read operation** for weight matrix
- Resonant adiabatic C-CIM: **resonant energy recycling** and **adiabatic charging** for FeCap array, *reducing dynamic power and resistive dissipation*



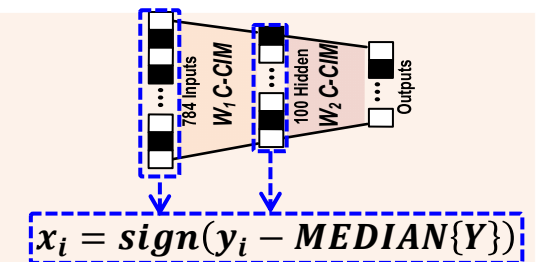
## **Experimental demonstration of FeCap-based RA C-CIM**

- RA C-CIM system based on cross-point FeCap array
- **RA weight writing and MAC computing** with reduced power consumption



## **FeCap-based RA C-CIM for edge AI application**

- **Co-optimizing the design and normalization algorithm**
- *High energy efficiency* for weight update and computation in edge AI

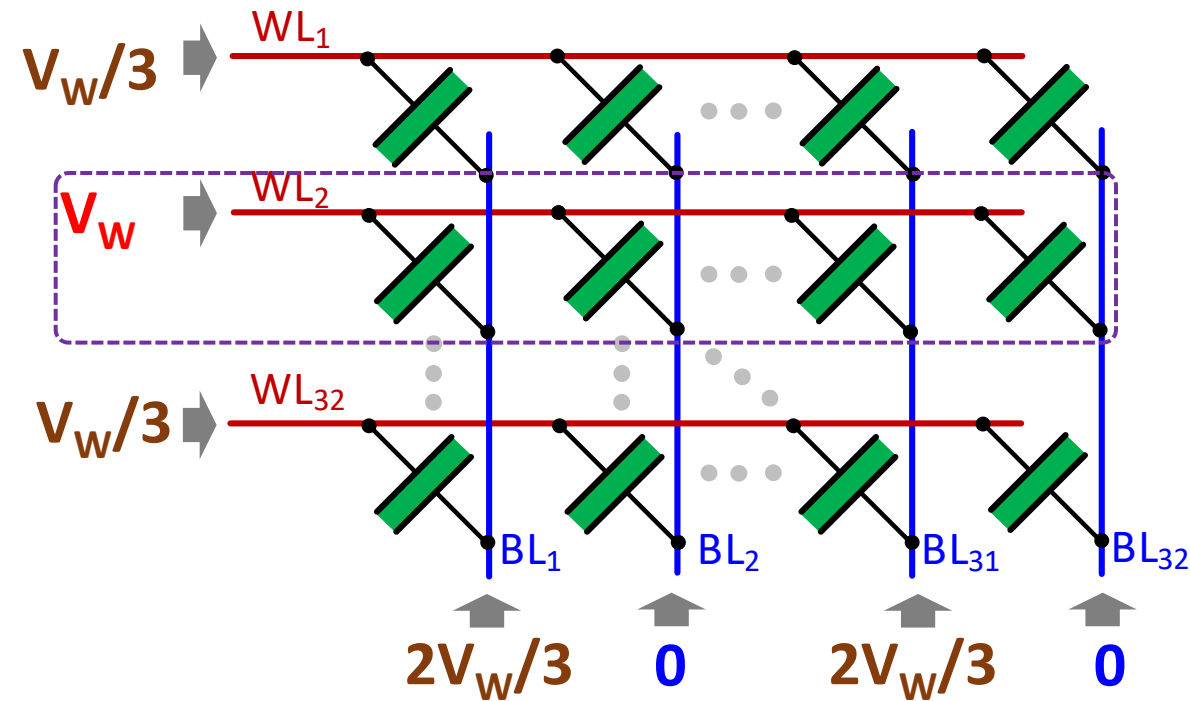
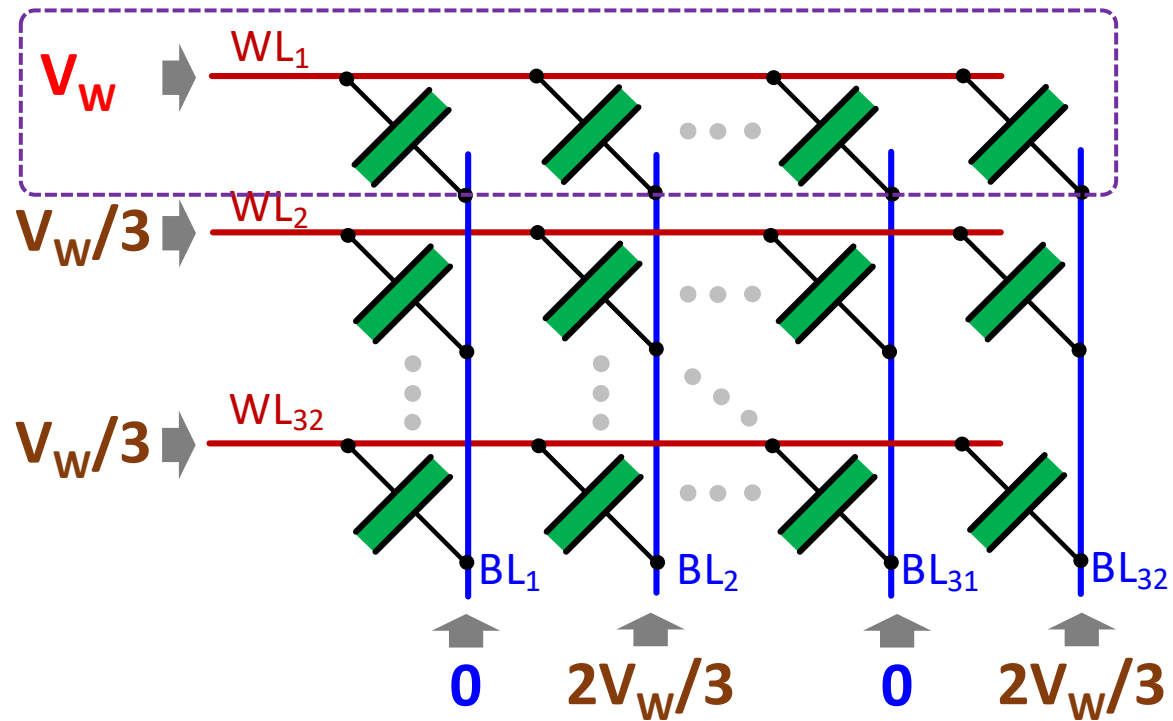


***Thank you for your attention!***

***Backup***

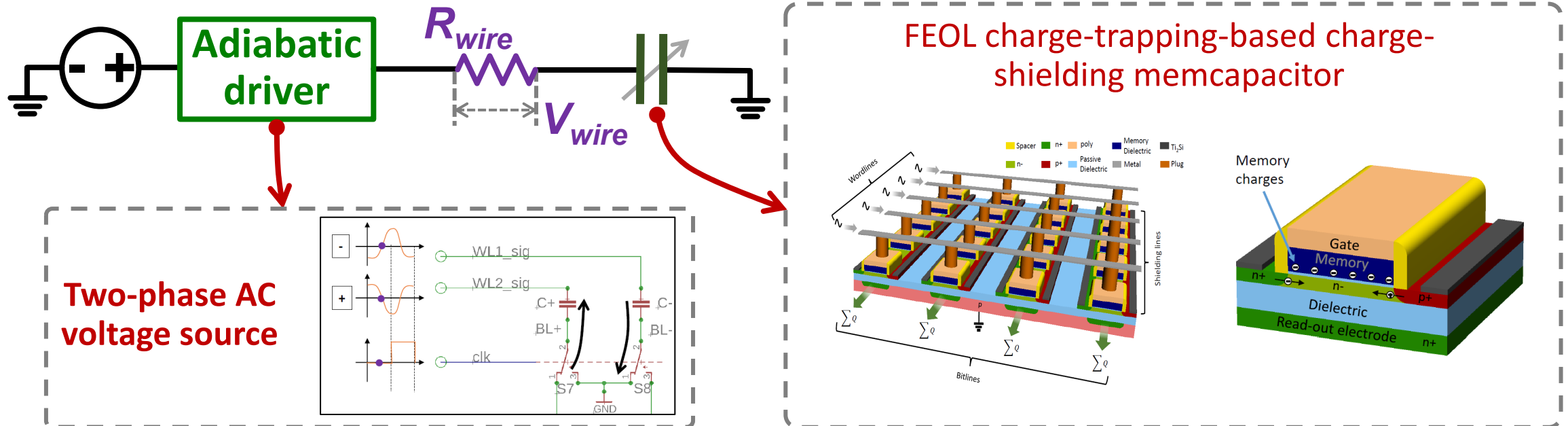
# Row-wise writing operation

- The  $V_w/3$  program scheme used in the experiment of this work



# Adiabatic technique-enhanced C-CIM

- Previous work: adiabatic C-CIM based on charge-trapping NVM<sup>[1]</sup>
    - Extra two-phase **AC power supply for adiabatic driver**
    - High write power for **charge-trapping-based capacitive NVM**
- } Suffer from additional area and energy cost



[1] Kai-Uwe Demasius, et al., Nat. Electron., 2021